



FABRIC MANUFACTURING II

Part: A

Course Code: FE 351-0723

General Terms Related to Knitting Technology

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TEXTILE

Textile

A term that originally applied only to *woven fabrics*. But the term textiles are now applied also to fibers, filament, yarn, natural and manufactured product.

Fabric

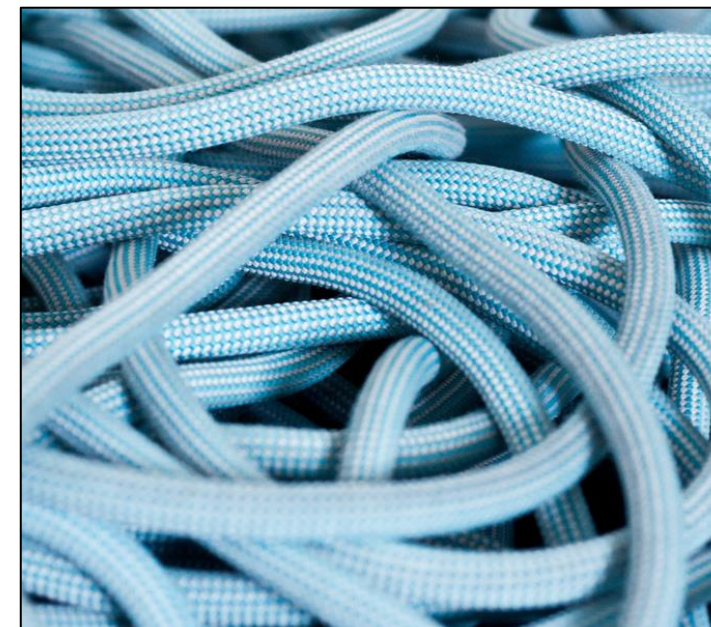
Fabric is a *flexible planar substance* constructed from *fibers, yarn* or *fabric* in any combination. Can also be produced directly from web of fiber by *bonding, fusing* or *interlocking*.



TEXTILE

Types of Fabric:

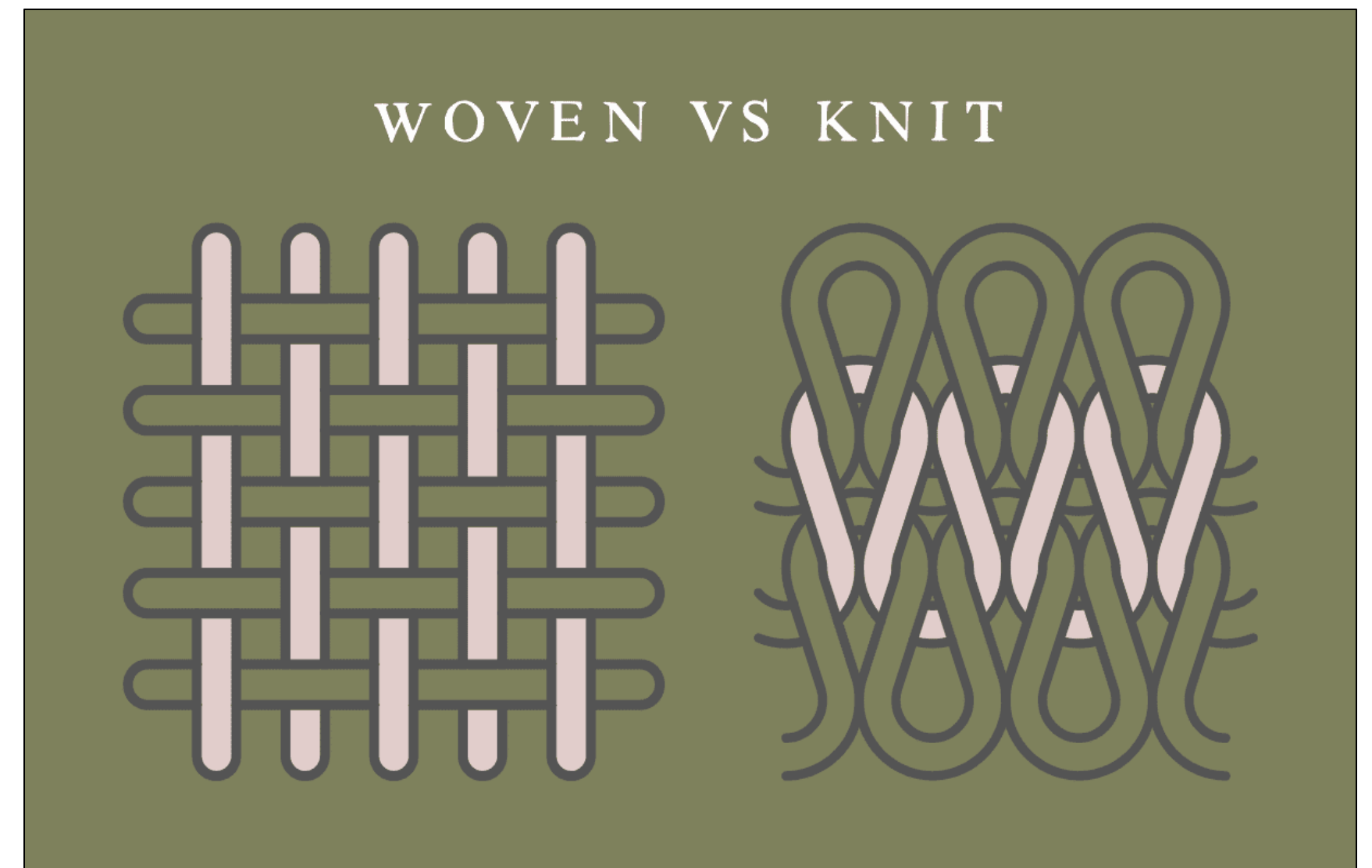
1. Woven Fabric: Produced by *interlacement* of two sets of yarn.
2. Knitted Fabric: Produced by *interloping* of one or one set of yarn.
3. Braided Fabric: Produced by *intertwining* of at least 3 groups of yarn from one set.
4. Non-woven Fabric: Flexible porous products consists of *one or more fiber layer* are bonded by chemical, thermal or mechanical processes.



WOVEN VS KNITTED FABRIC

Knitted fabric differs vastly from woven fabrics. Woven fabric is formed substantially by *interlacing* of a series of lengthwise and crosswise threads.

Knitting is the method of creating fabric by *transforming continuous strands of yarn into a series of interlocking loops*, each row of such loops hanging from the one immediately preceding it.



HISTORICAL BACKGROUND OF KNITTING TECHNOLOGY

First hand operated knitting machine invented in 16th Century by an English man - *William Lee (1589)*. A skilled hand knitter produce *120-150 stitches per minute*, whereas a modern high speed circular knitting machine makes *about 20M* stitches per minute.

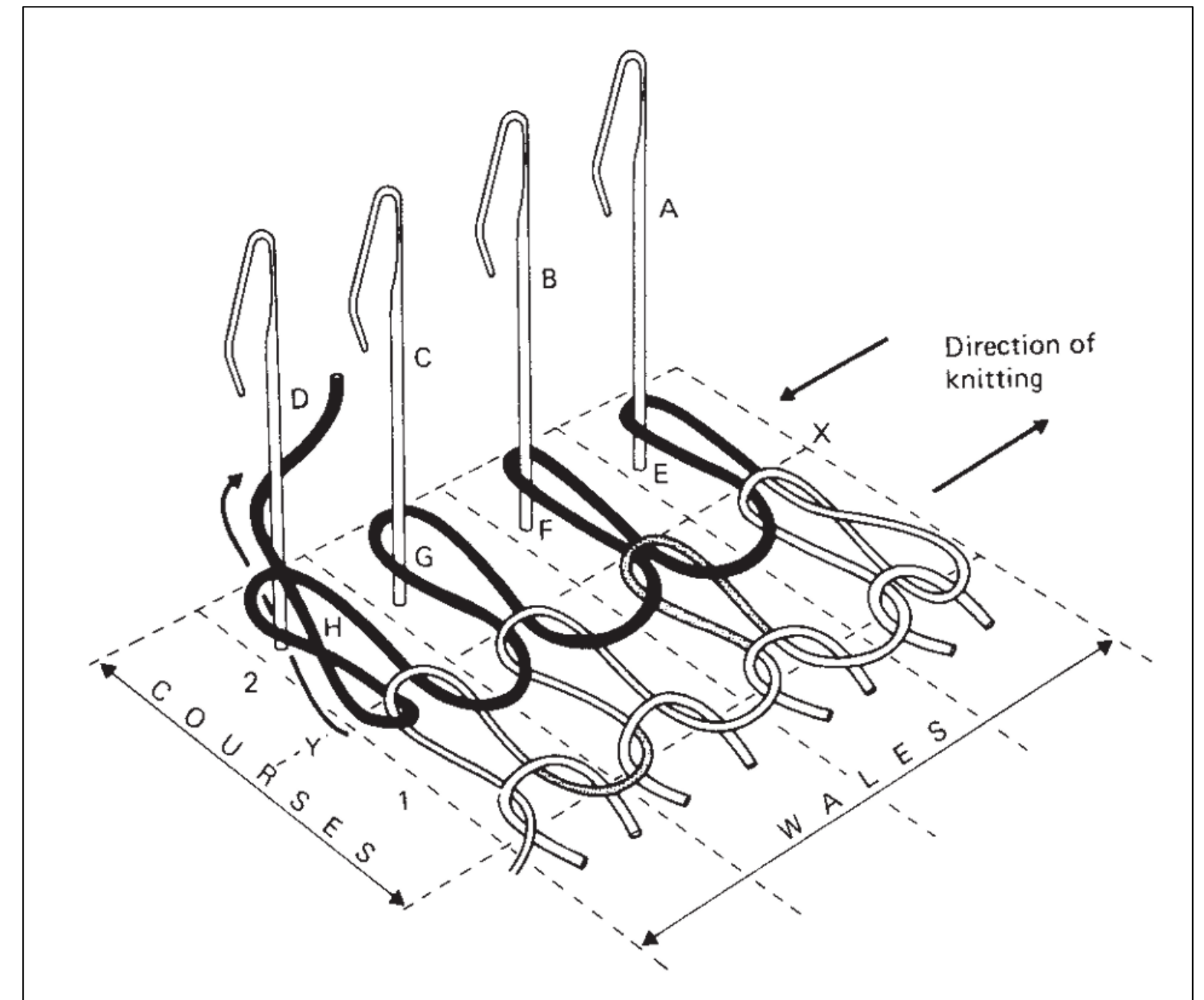
- * *Pins* replaced by *Needles*
- * *Hand* replaced by *Needle Holder*
- * *Finger* replaced by *Yarn Feeder*
- * *Yarn Ball* replaced by *Yarn Packages*



TYPES OF KNITTING

Weft Knitting:

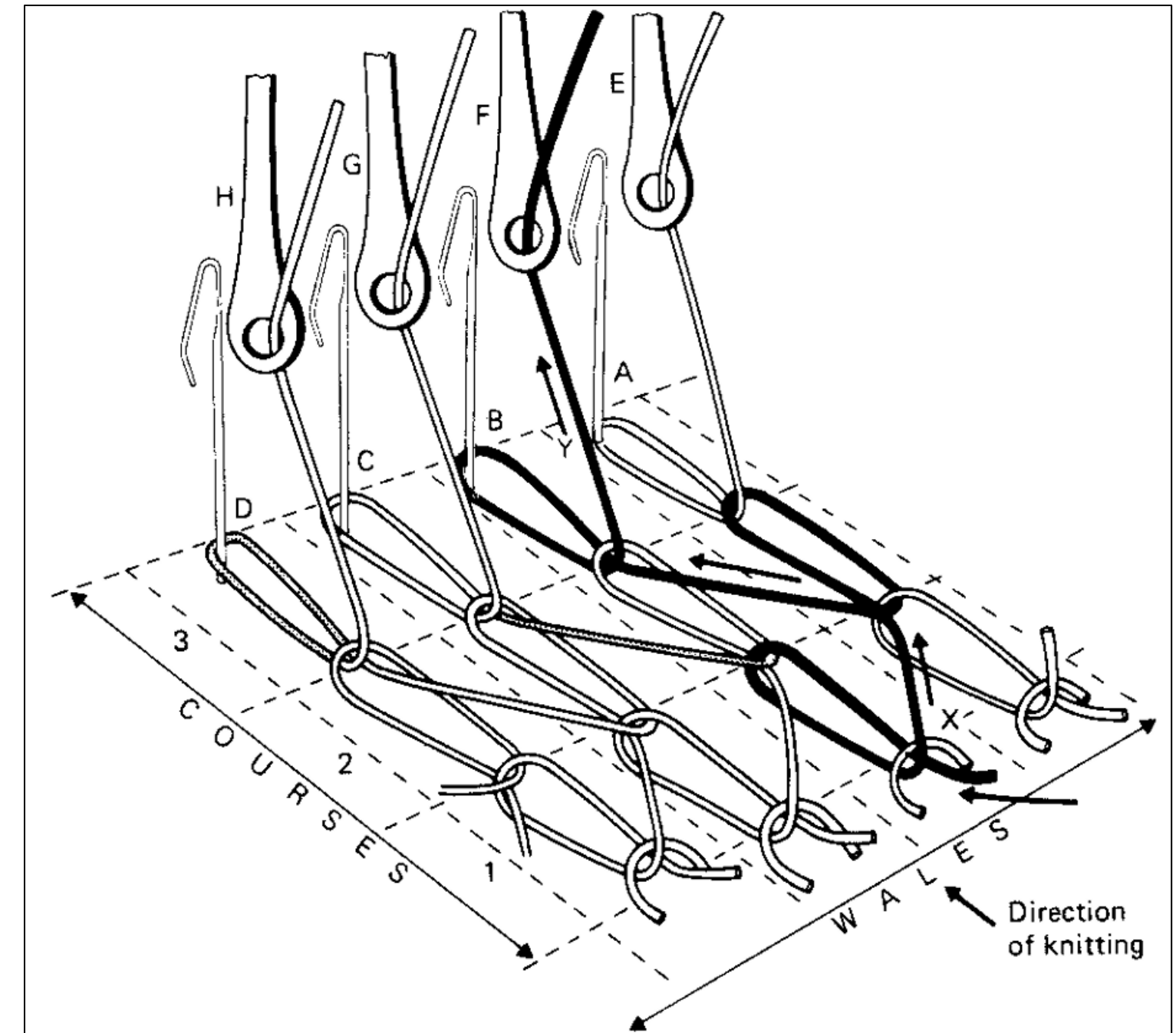
A weft knitted structure is a *horizontal row of loops can be made using one thread* and *thread runs in horizontal direction*.



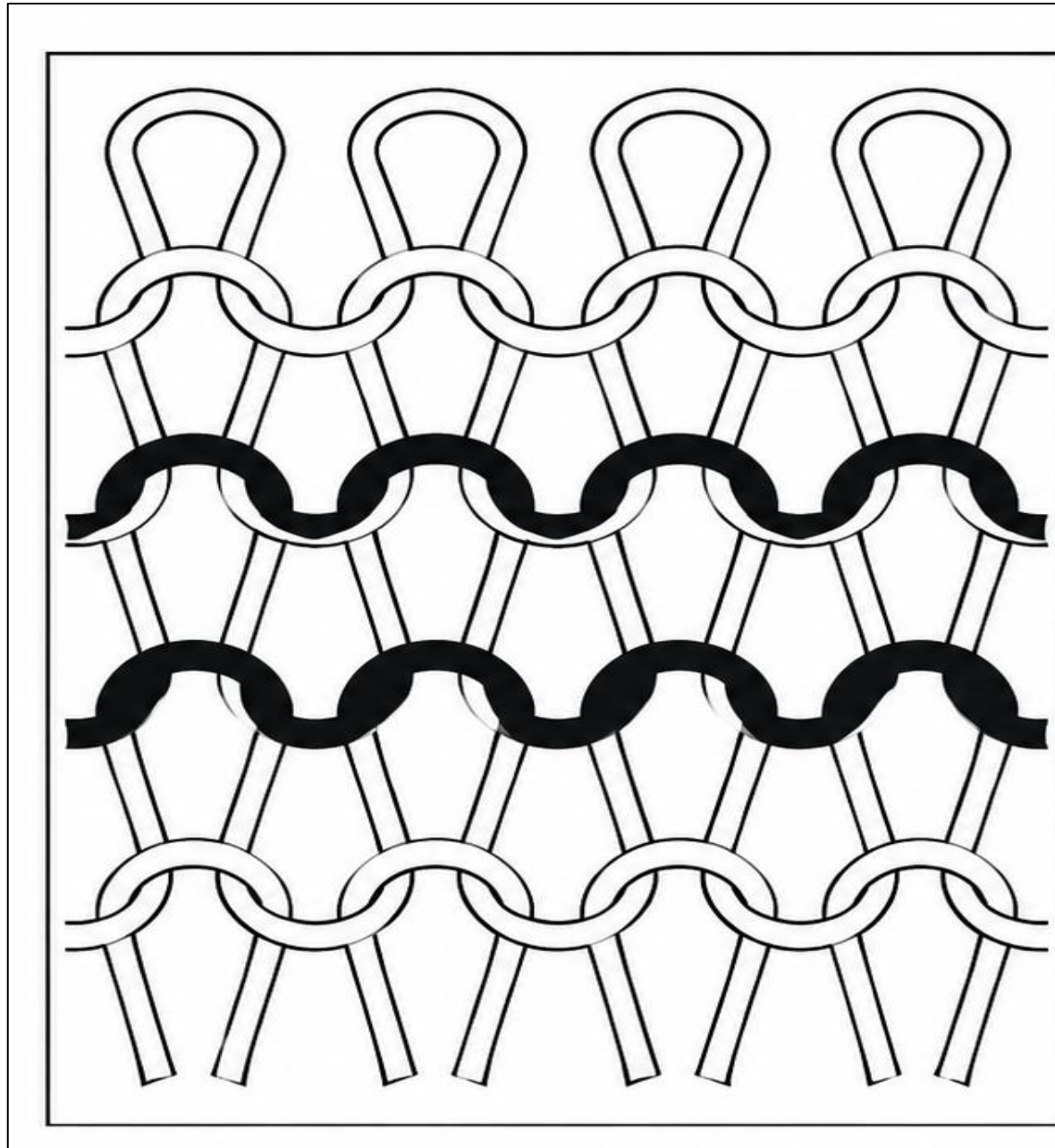
TYPES OF KNITTING

Warp Knitting:

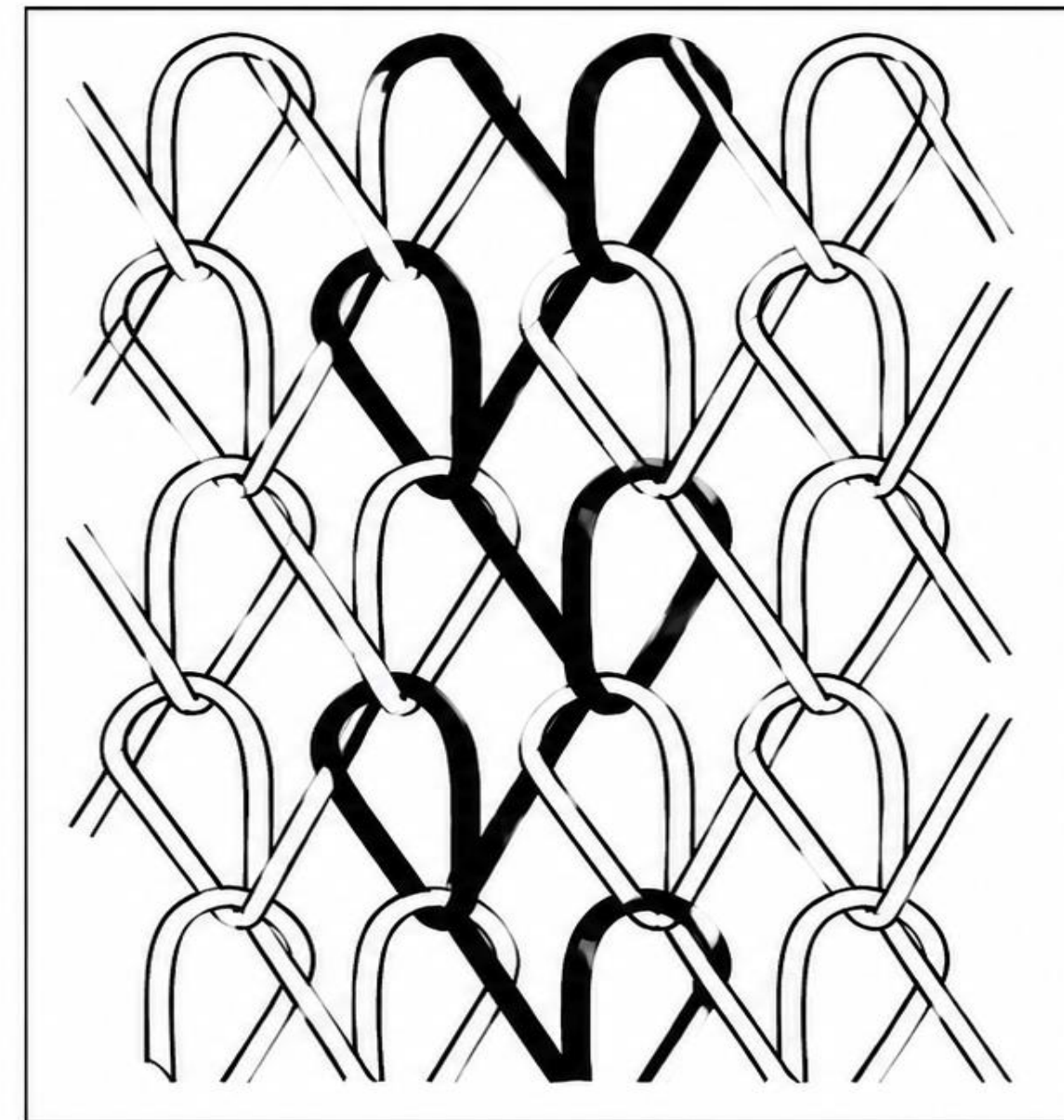
In warp knitted structure *each loop in horizontal direction is made from a different thread* and the number of threads used to produce such fabric *is equal to the number of loops in a horizontal row.*



TYPES OF KNITTING (CONTINUED)



WEFT KNITTED
STRUCTURE

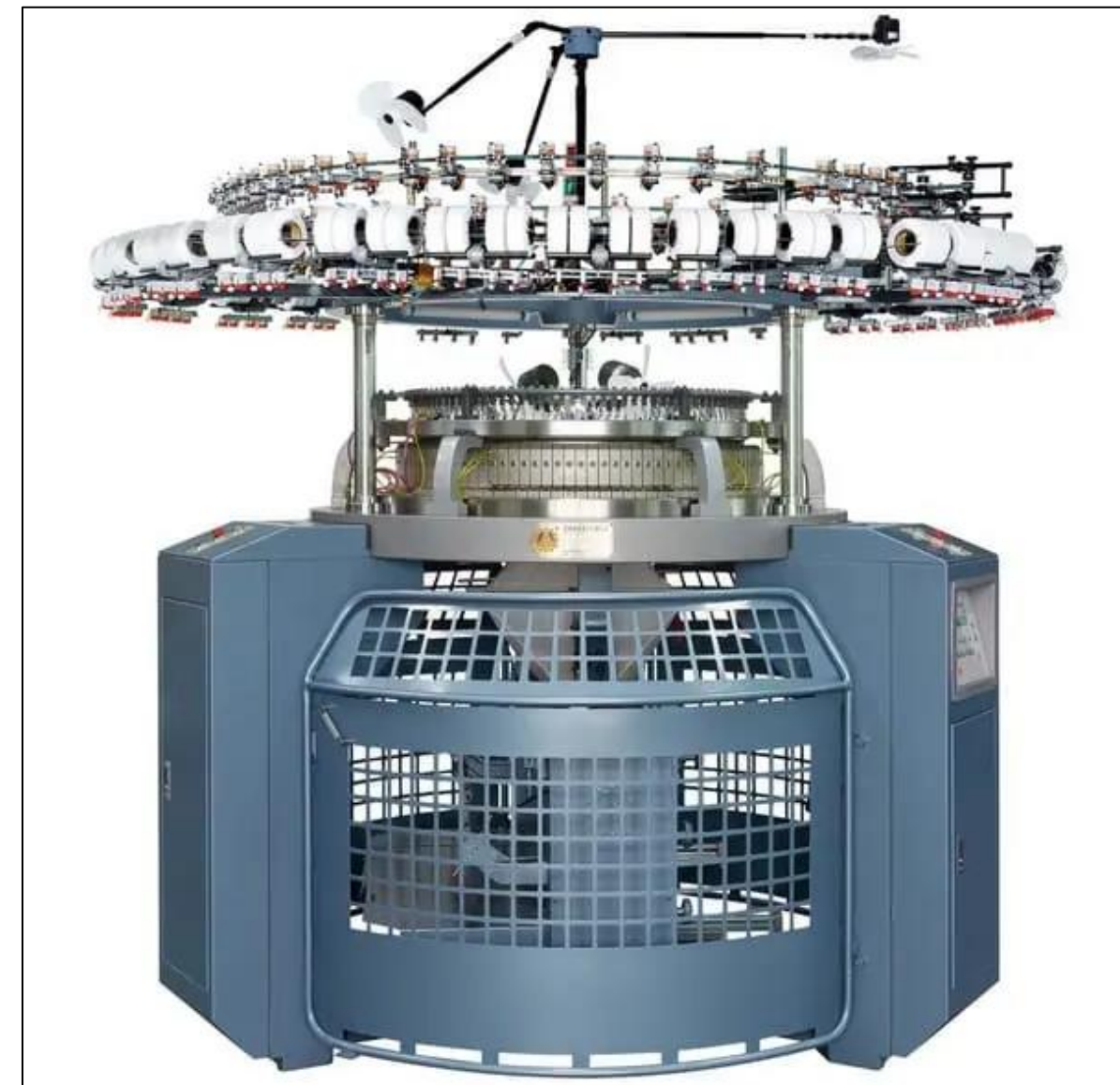


WARP KNITTED
STRUCTURE

TYPES OF KNITTING (CONTINUED)



TRICOT WARP KNITTING MACHINE



SINGLE JERSEY CIRCULAR KNITTING MACHINE

TYPES OF KNITTING (CONTINUED)

Warp Knitting	Weft Knitting
1. Loop formation in the fabric takes place <i>wale wise</i> .	1. Loop formation takes place <i>course wise</i> .
2. Yarns are supplied from a <i>beam</i> in the machine.	2. Yarns are supplied from <i>cones</i> in the machine.
3. Yarns produced from <i>synthetic fibre</i> are mostly used.	3. Yarns produced from <i>natural fibre</i> are mostly used.
4. <i>Antistatic lubricants</i> is used to prevent <i>static electricity</i> buildup (a big issue since many yarns run in parallel at high speeds).	4. Special <i>low-viscosity knitting oil</i> is used which reduces friction between yarn, needles, and machine parts.
5. Design change process (link in the <i>patterned drum or pattern disc</i>) of the machine is less time consuming.	5. Design change (<i>needle and cam</i>) process of the machine is more time consuming.

TYPES OF KNITTING (CONTINUED)

Warp Knitting	Weft Knitting
6. Knitting elements of the machine <i>act unitely</i> .	6. Knitting elements <i>act independently</i> .
7. Fabrics have unlimited application, such as inner and outer wears, industrial, technical, and domestic.	7. Limited application, mostly apparel.
8. Latch, bearded or compound needles are used in warp knitting machine.	8. Usually latch needles are used on weft knitting machines.
9. Both Fabrics and machines are <i>expensive</i> .	9. These are <i>cheaper</i> than warp knitting.
10. <i>All the loops</i> of a course are formed <i>exactly at the same time</i> .	10. The loops of a course are formed <i>sequentially</i> .

TYPES OF KNITTING (CONTINUED)

Warp Knitting	Weft Knitting
11. <i>More</i> stable fabric dimensionally.	11. <i>Less</i> stable.
12. The fabric is <i>more suitable</i> for cutting and sewing.	12. <i>Less</i> suitable.
13. <i>Every needle</i> is required to have <i>individual yarn feed</i> .	13. <i>A single yarn is sufficient for all the needles.</i>

KINK OF YARN, KNITTED LOOP, KNITTED STITCH

Kink of Yarn

A length of yarn that has been *bent into a shape appropriate* for its transformation into a weft knitted loop.

Knitted Loop

A kink of yarn that is *intermeshed at its base* i.e. when *intermeshed two kink of yarn* is called loop.

Knitted Stitch

Basic unit of intermeshing and usually consists of *three or more intermeshed loops*.



KINK OF YARN, KNITTED LOOP, KNITTED STITCH

The basic element of a weft knitted structure is Loop or Stitch. The fabric may be composed of *Knit, Tuck and Miss loops* but *without knit loop, there can't be a knitted fabric*.

When a fabric is composed of *all knit loops*, it is called *Plain fabric*. Example: Plain Single Jersey, Plain Interlock, Plain Rib or 1x1 Rib. When a fabric is composed of *Knit along with Tuck and or Miss loops*, it is called *Derivative*.

TOP ARC, BOTTOM ARC, LEGS

Top Arc

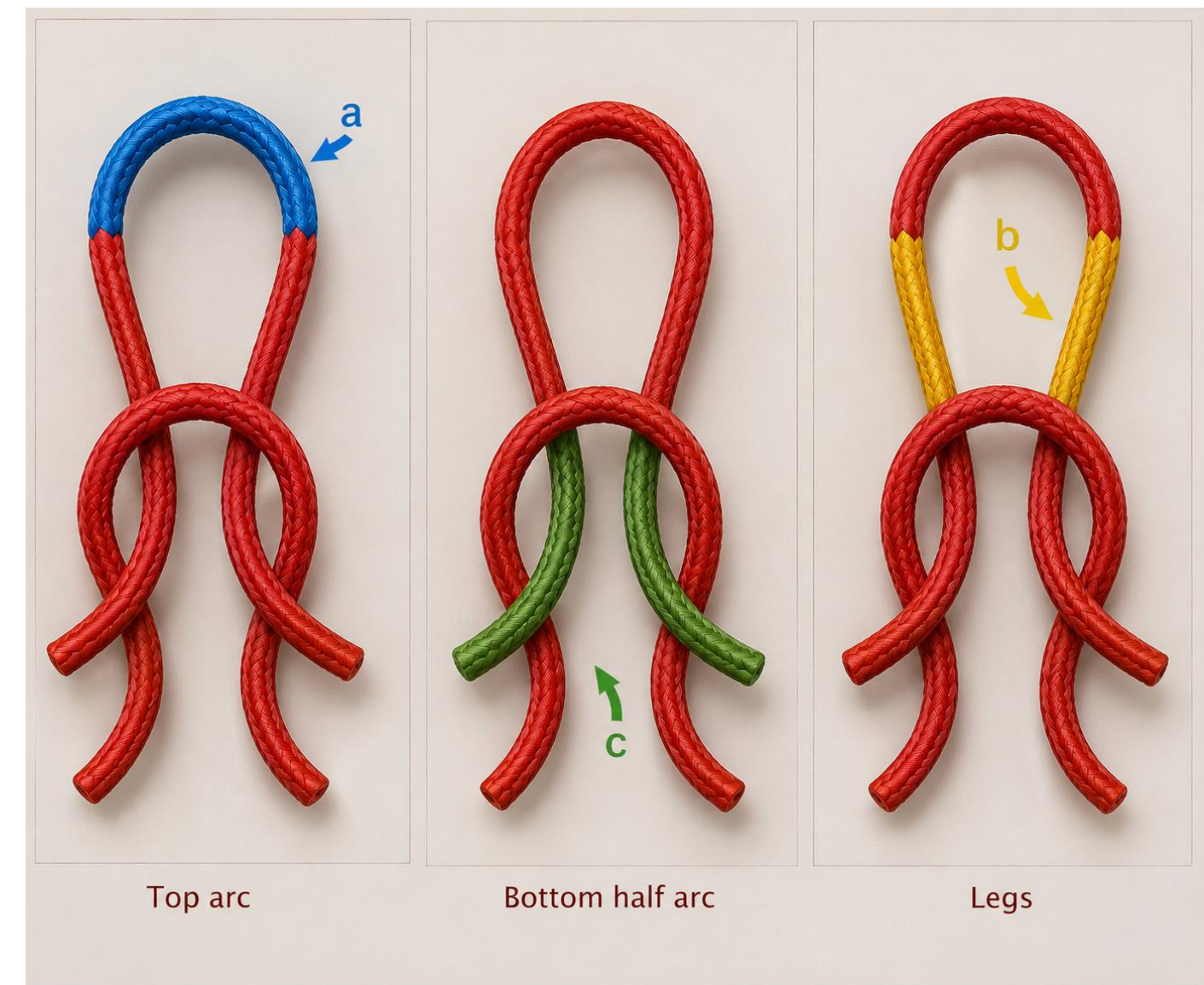
The *upper curved portion* of the knitted loop is called top arc.

Bottom Half-arc

The *lower curved portion* that constitutes in a weft knitted loop, *half of the connection to the adjacent loop* in the same course.

Legs or side limbs

The *lateral parts of the knitted loop* that connect the top arc to the bottom half-arcs.



NEEDLE LOOP, SINKER LOOP

Needle loop

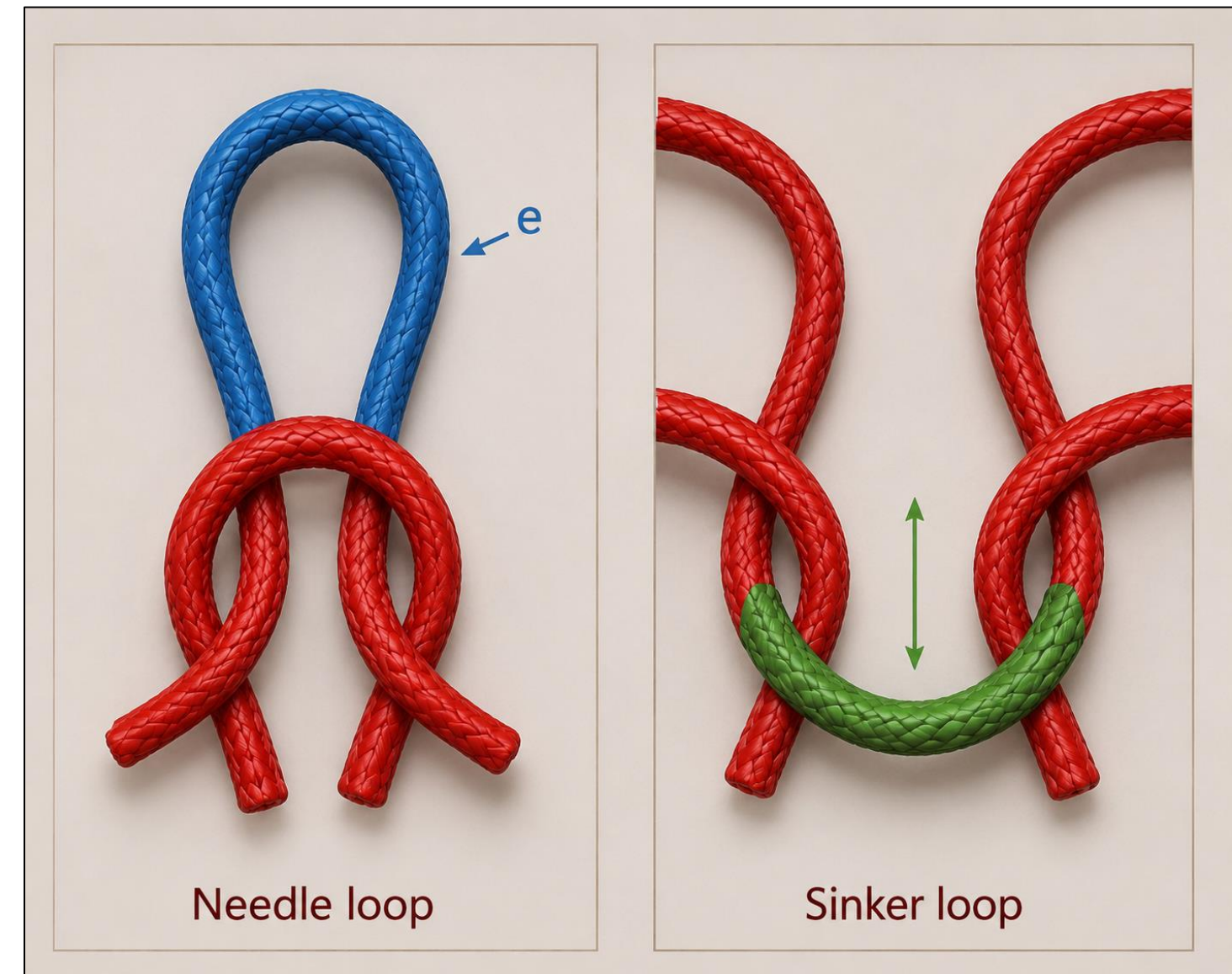
Needle loop formed by the top arc and the two legs of the weft knitted loop. **Needle loop** is the upper part of the loop produced by **the needle** drawing the yarn.

Needle loop = Top arc + Two legs

Sinker loop

The yarn portion that connects two adjacent needle loops belonging in the same knitted course. Bottom arc also called sinker loop.

The size of the sinker loop can be altered by the action of sinker.



KNITTED LOOP STRUCTURE, COURSE, WALES

Knitted loop structure

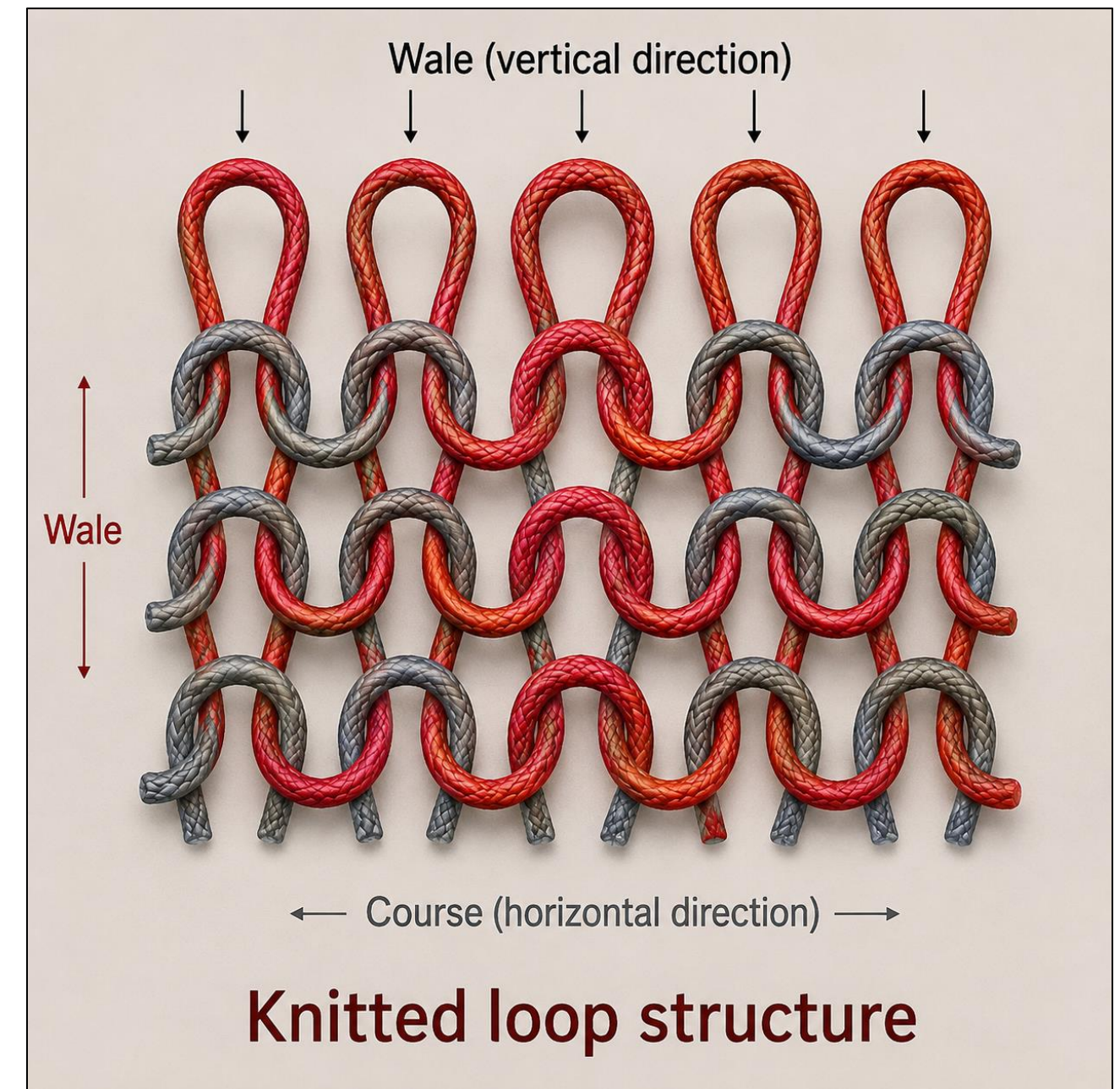
The properties of a knitted structure are largely determined by *the interdependence of each stitch* with its neighbors on either side and above and below it.

Course

A course is a *predominantly horizontal row of loops produced by adjacent needles* during knitting cycle.

Wales

A wale is a *predominantly vertical column of needle loops produced by same needle* at successive knitting cycles.



STITCH LENGTH

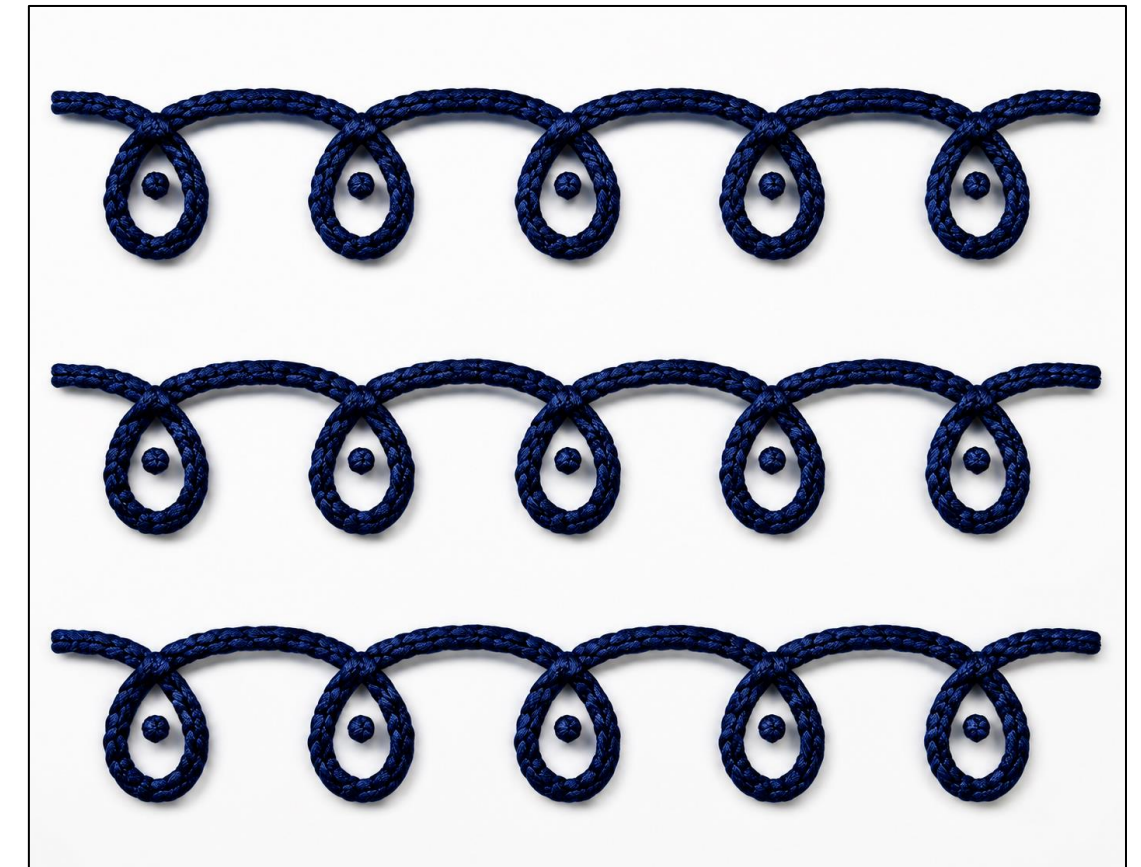
Stitch Length

The length of yarn required to produce a complete knitted loop (i.e Needle loop and sinker loop) is known as stitch length or loop length.

Depending upon the type of stitch or loop combination used in a course, individual stitch length can vary.

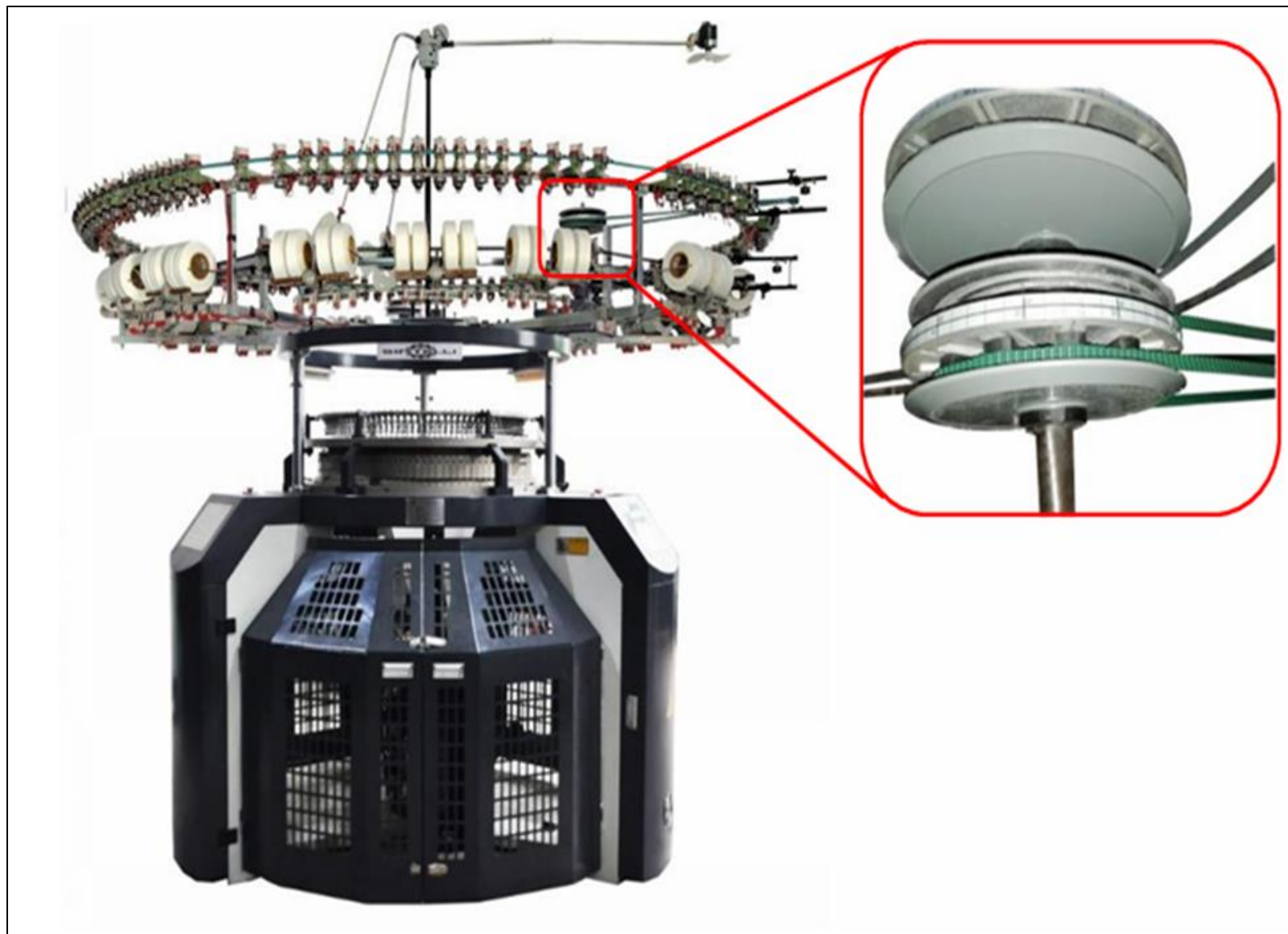
Therefore, *stitch length is always calculated as an average of a number of loops in a course.*

Generally, the *larger the stitch length*, the more extensible and *lighter the fabric* and the *poorer the cover and bursting strength.*

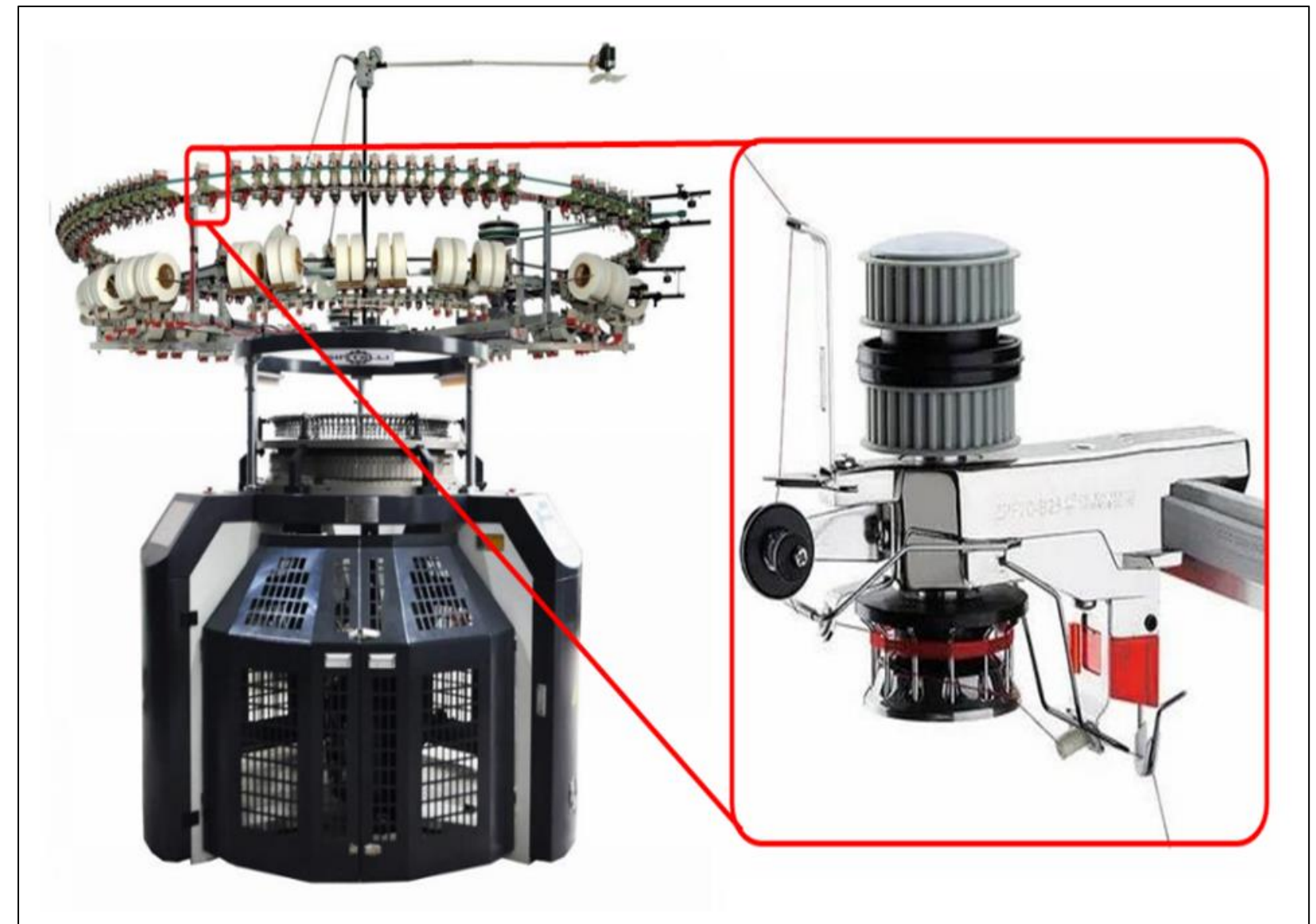


STITCH LENGTH

Stitch Length Controlling Parts



VDQ PULLEY

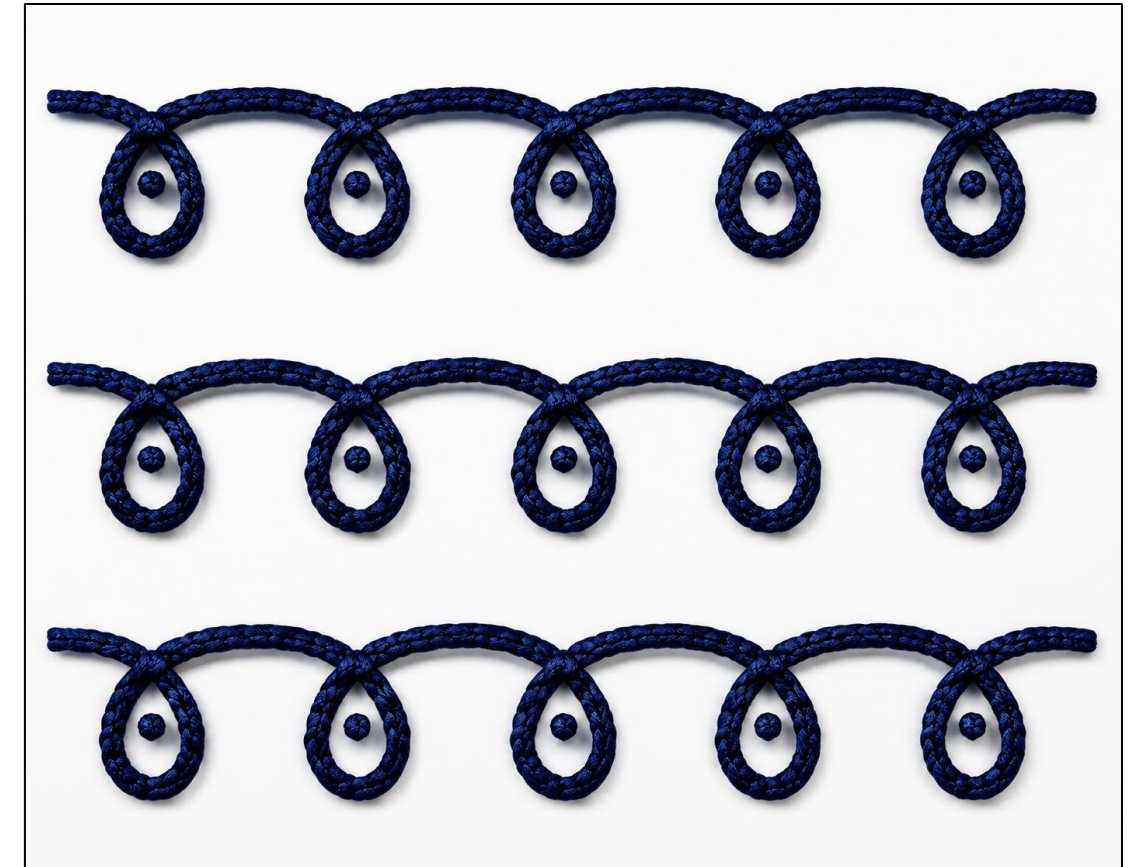


POSITIVE FEEDER

COURSE LENGTH

The *length of yarn required* to produce a complete knitted course is known as course length.

$$\begin{aligned}\text{Course length} &= \text{No. of loops per course} \times \text{Stitch length} \\ &= \text{No. of needles} \times \text{Stitch length}\end{aligned}$$



STITCH DENSITY

Stitch Density

Total number of needle loops in a square area such as - Square inch or Square centimeter.

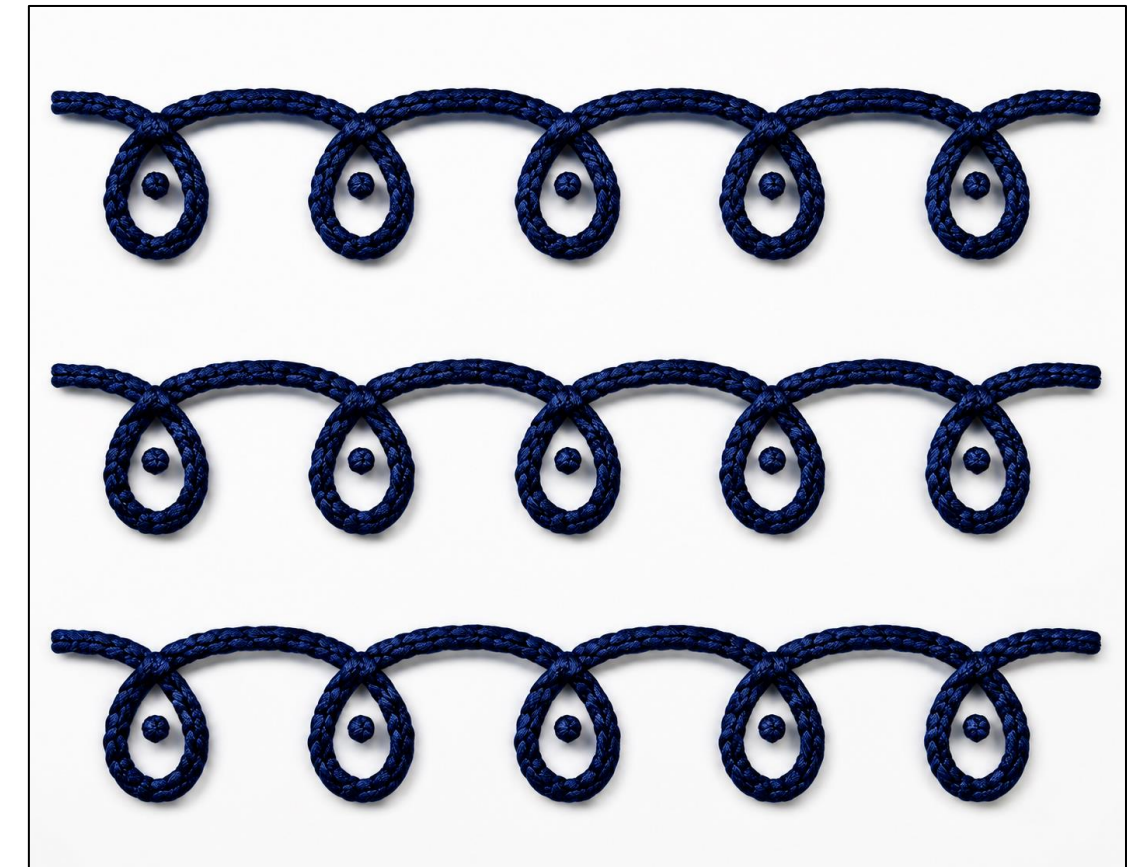
Stitch Density (SD) = Wales per inch (WPI) x Courses per inch (CPI)

- Higher the stitch density heavier is the fabric (High GSM).
- Since knit fabric is stretchable, the commercial practice for trading knit fabric is by weight.

* Say, WPI = 5 & CPI = 3

So, $SD = 5 \times 3 = 15$

There are 15 loops per square inch area.



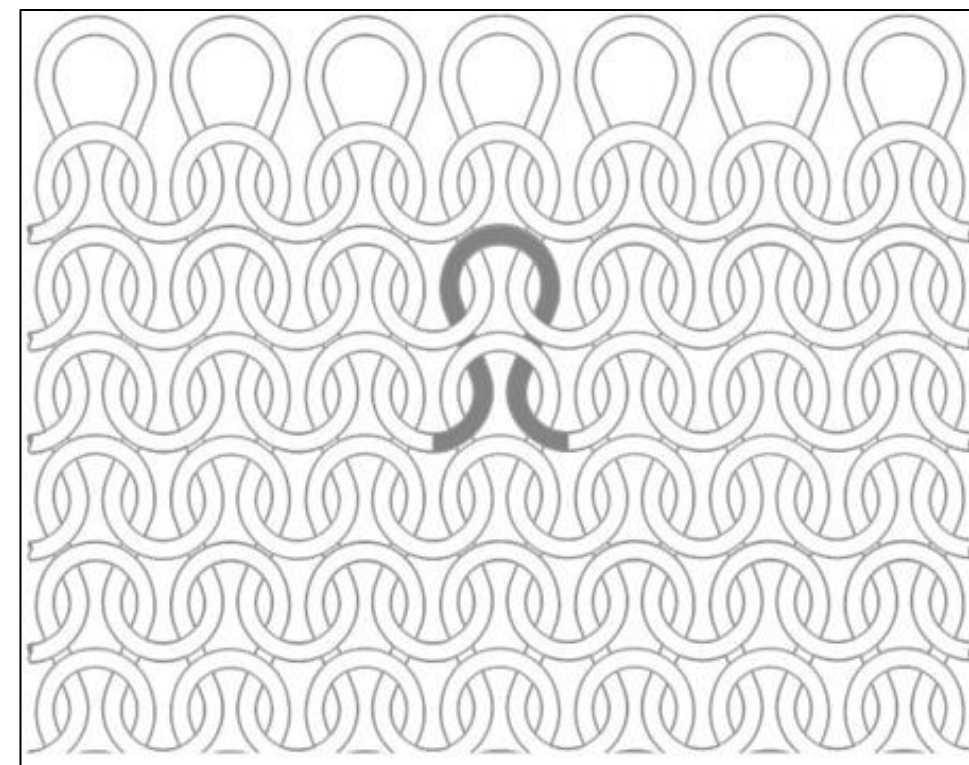
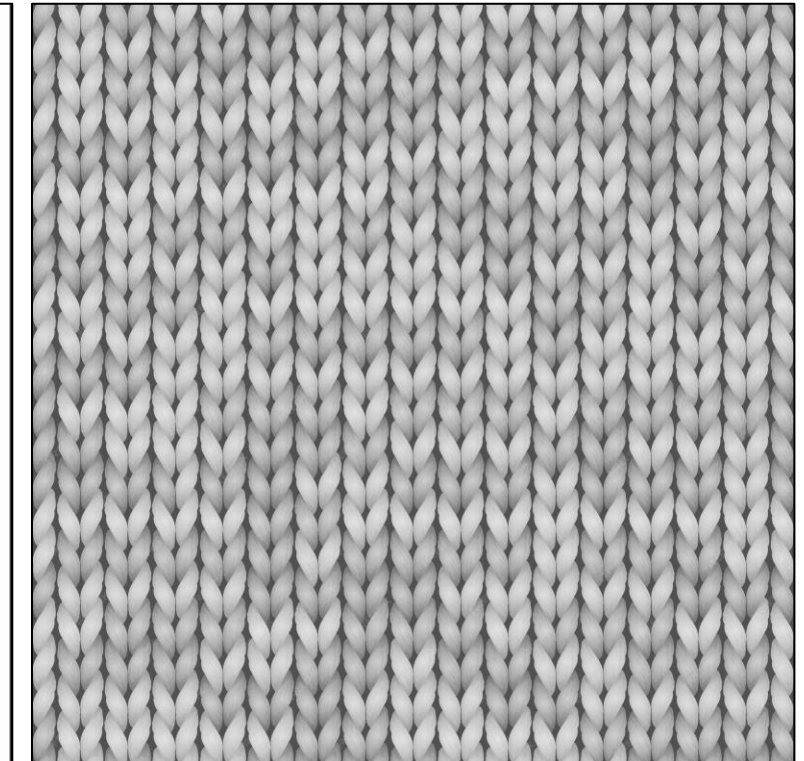
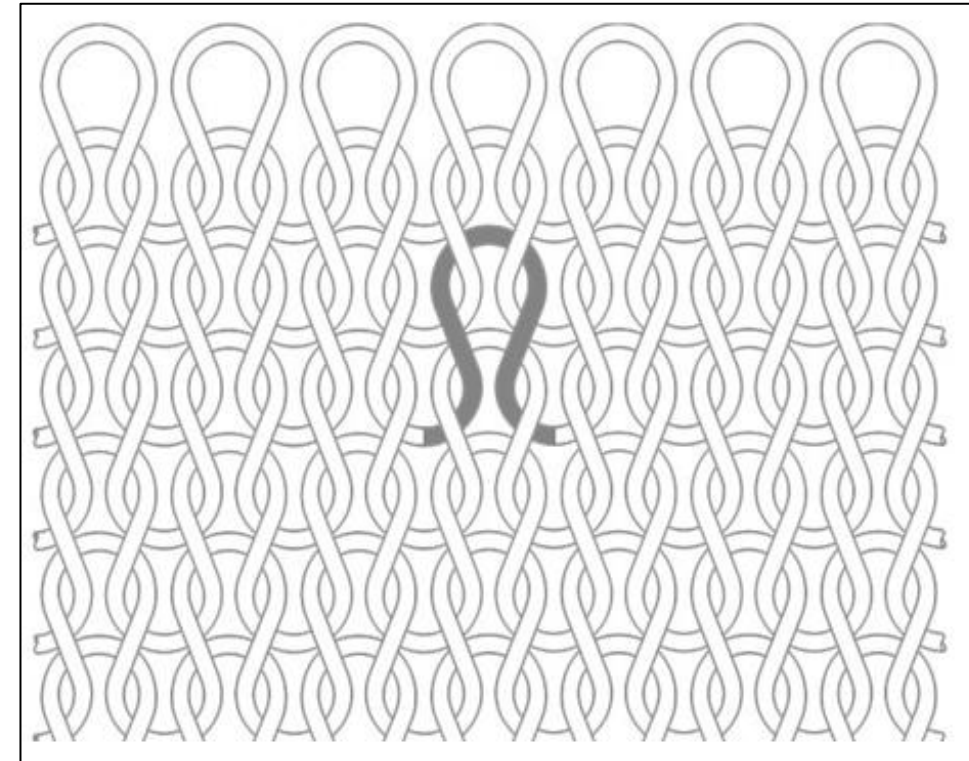
FACE LOOP, BACK LOOP

Face Loop

A loop that is drawn from the previous loop and intermeshed with the succeeding loop in such a way that its *head lies below the legs of new loop and its legs lie above the head of the preceding loop* is called Face Loop.

Back Loop

A loop that is intermeshed in such a way that its *top arc and the bottom arcs are situated above the legs* of the stitch formed in the same wale in the previous and in the following course is called Back Loop. It is also known as reverse loop or purl stitch.



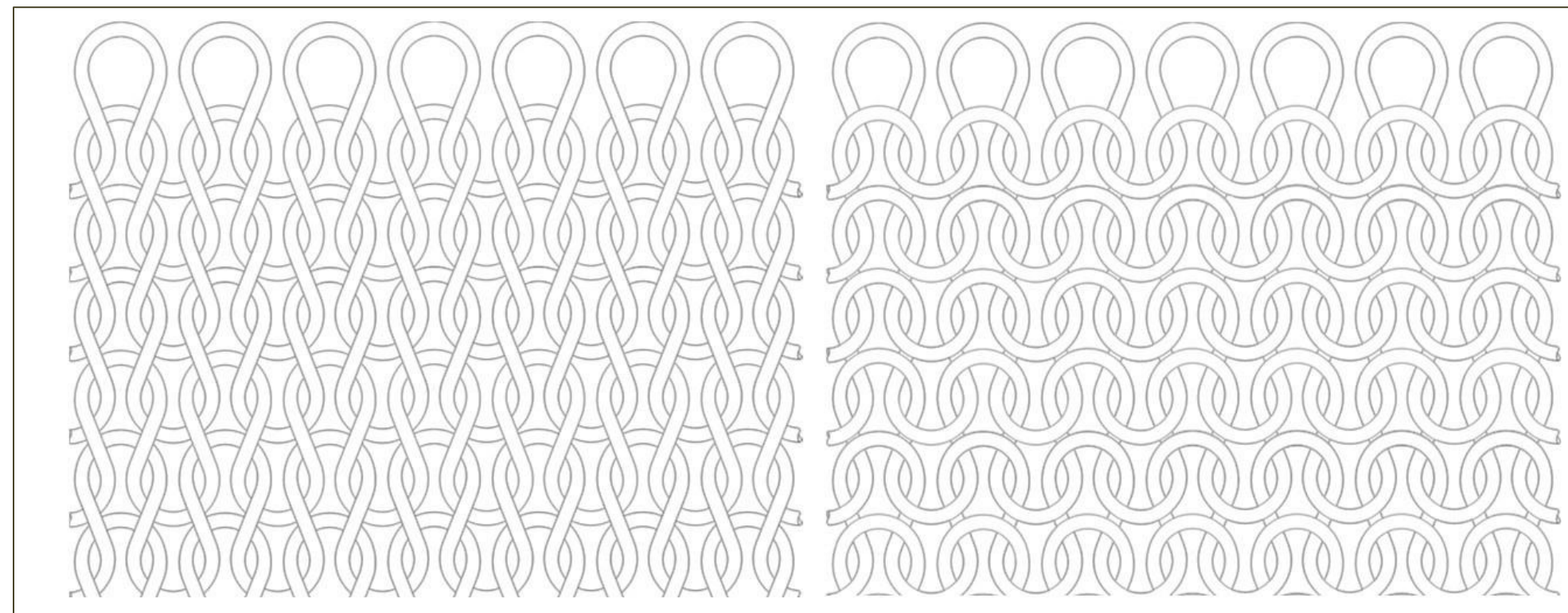
TECHNICAL FACE, TECHNICAL BACK

Technical Face

The surface of the fabric where *legs of loops are dominantly visible* is called Technical Face of the fabric. Generally, it is the outer surface of the fabric when producing. Wales in the form of vertical lines are prominent on this side.

Technical Back

The side of the fabric which *contains all back loops or purl loops* is known as technical back. Generally, it is the inner surface of the fabric when producing.



FABRIC STRUCTURE

Single Face Structures

Single face structures are made using *only one set of needles in both warp and weft knitting*. The needles may be arranged in a *straight line or in a circle*. All adjacent *needles have their hooks facing the same direction*. The needle heads pull the new loops down through the old loops in the same direction.

Double Face Structures

Double face structures are produced by using two sets of independently controlled needles in warp and weft knitting. The *hooks of one needle set face in the opposite direction to the other needle set*. Both sets of needles pull loops from the same yarn but in opposite directions. The *fabric is formed in the gap between the two needle sets. One side of the fabric shows the face loops of one needle set, and the other side shows the face loops of the second needle set. The two fabric faces are connected by sinker loops or underlaps inside the fabric*. The reverse stitches remain inside the structure and are mostly hidden.

STITCH DENSITY

Balanced Face Structure

Double face structure *has equal stitches on both needle beds*. Both fabric surfaces show the same stitch arrangement, making the fabric stable and preventing edge curling.

Fabric Drawn Off

- ❖ Fabric is always *pulled away from the needles on the side opposite to the needle hooks*.
- ❖ When two sets of needles are used, they can be *placed vertically or at an angle to each other*.
- ❖ The hooks of both *needle sets face away from each other*.
- ❖ The fabric is formed and *pulled out through the space between the two needle sets*.

NEEDLE CARRIER

Needle

Principal knitting element.

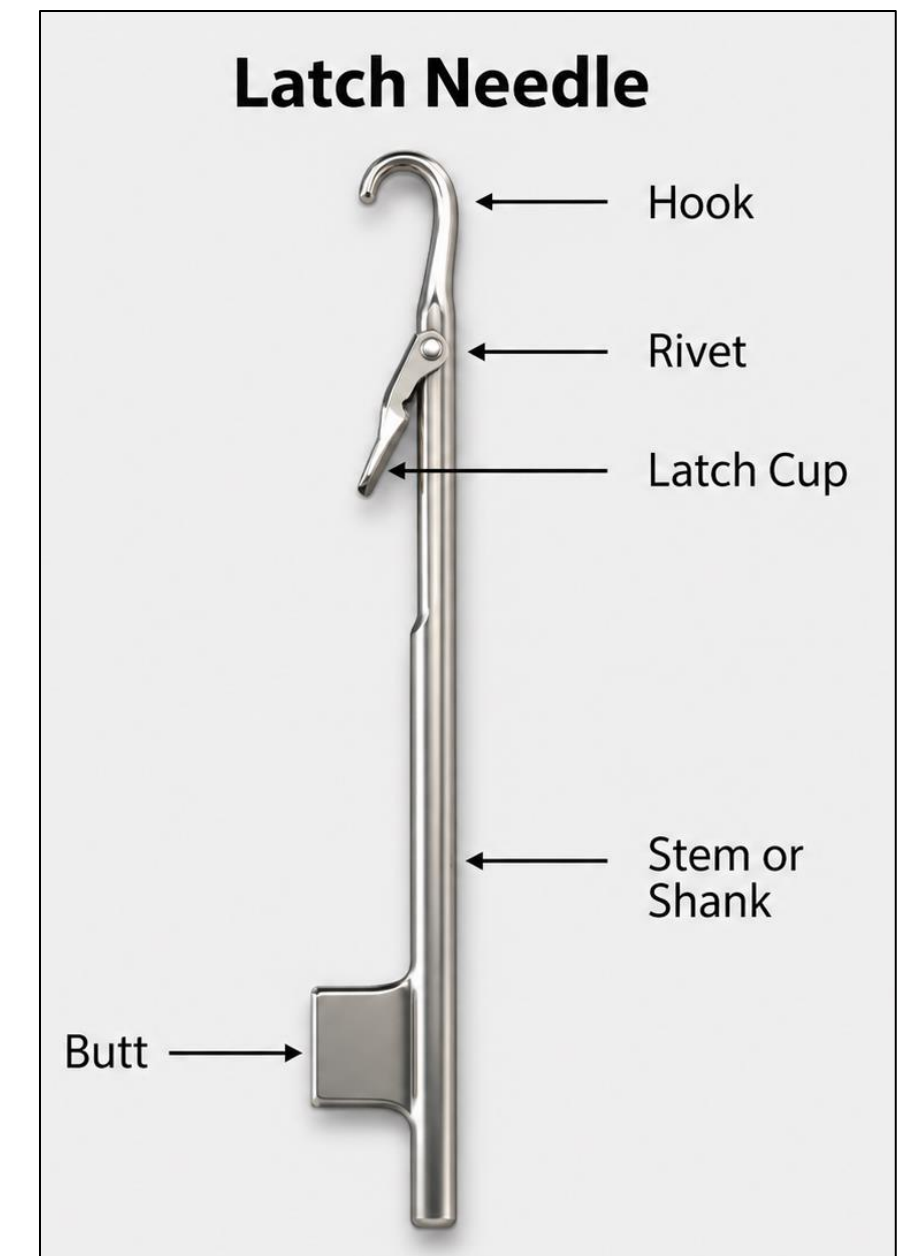
1. Independent Needles: Can slide in the needle bed & *can be moved individually during stitch forming*.
2. United Needles: *Fixed in needle bar* & can be moved in unison during knitting process.

Needle Carrier

Needle bed or needle carrier is the place where the needles are located or mounted in a knitting machine. *Needle moves up and down in the track of a needle carrier*. Two types: 1. Cylindrical or circular 2. Flat.

The Working Diameter

Two opposite needles in a circular bar, expressed in inch.



CYLINDER, DIAL

The Working Width

The working width of the machine should not be miss used with the "actual working width". *The working width of a knitting machine is a measure (inch or inches) describing the distance between the first and last needle in a flat needle bed, whilst the "actual working width" is the distance between the first and last needle actually used in the needle bed during knitting.*

Cylinder

Cylinder is a circular steel bed *having grooves/tricks/cuts* on its *outer periphery* into which the needles are mounted. With reference to the tricks, the needles move vertically up and down by their butt being in contact with the cam track. The diameter of the cylinder also varied based on the type and width of the fabric.

Dial

Dial is the upper steel bed used in double knit circular machines. Into the grooves of the dial, the *needles are mounted horizontally* and are allowed to *move radically in and out by their dial cams*. The *number of grooves per unit space conforms with the cylinder gauge in most of the cases.*

KNITTING MACHINES

The machines those produced fabrics by intermeshing or interloping of one or one set of yarn. Knitting machines are divided as follows:

Weft Knitting Machine

A machine producing a knitted fabric by intermeshing loops formed *successively across the width of the fabric* from a yarn fed substantially crosswise to the length of fabric.

Warp Knitting Machine

A machine in which the fabric formation occurs by interlacement of loops formed *simultaneously across the full width of the fabric* from a system of yarns which are fed to the needles *substantially in the direction of the length of fabric*.

Other loop forming and combined technique machines

Machines that form the fabric by the technique of intermeshing of yarn loops but *which cannot be defined either as weft or as warp knitting machines*.

MACHINE PARTS



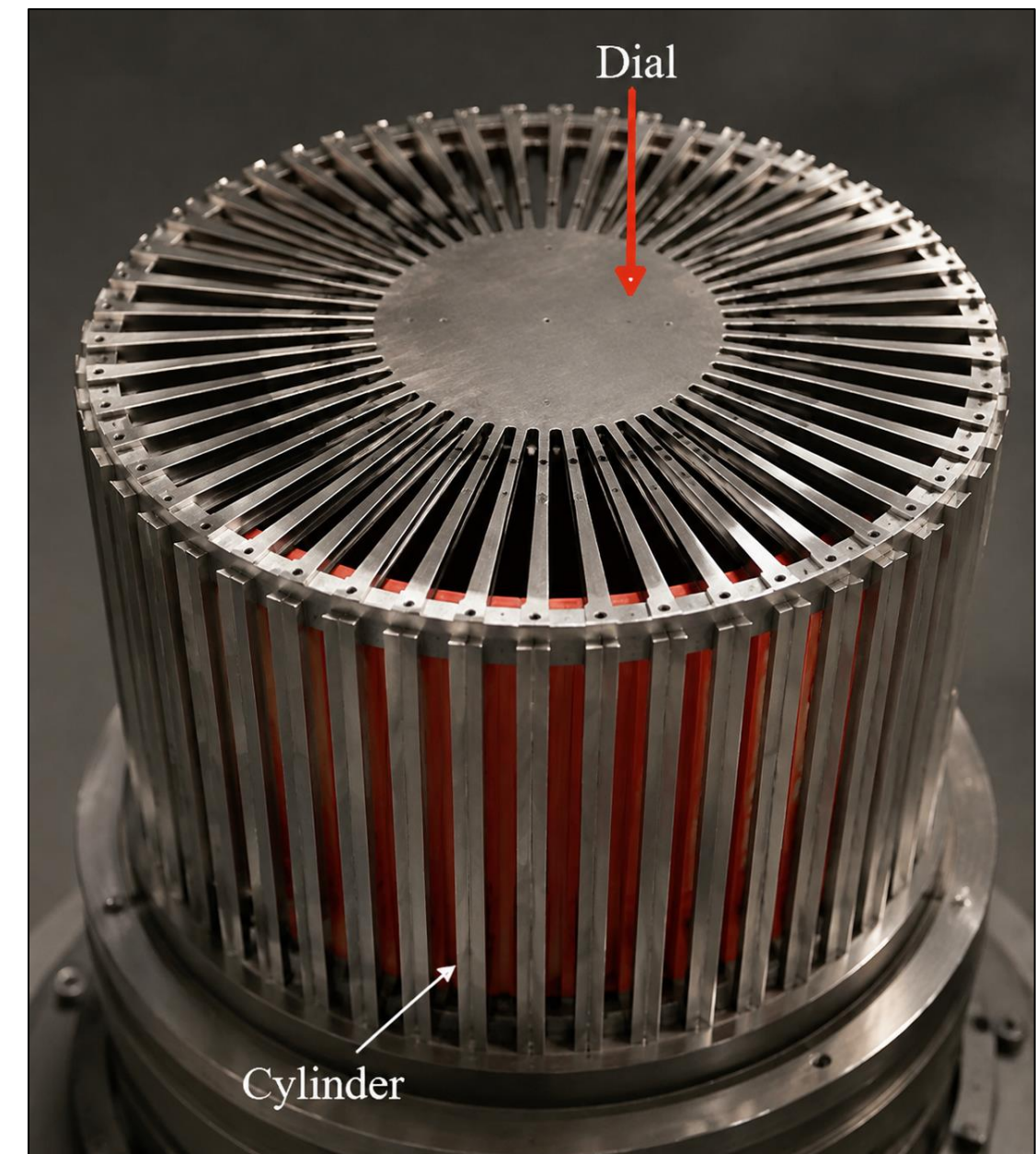
Needle carrier



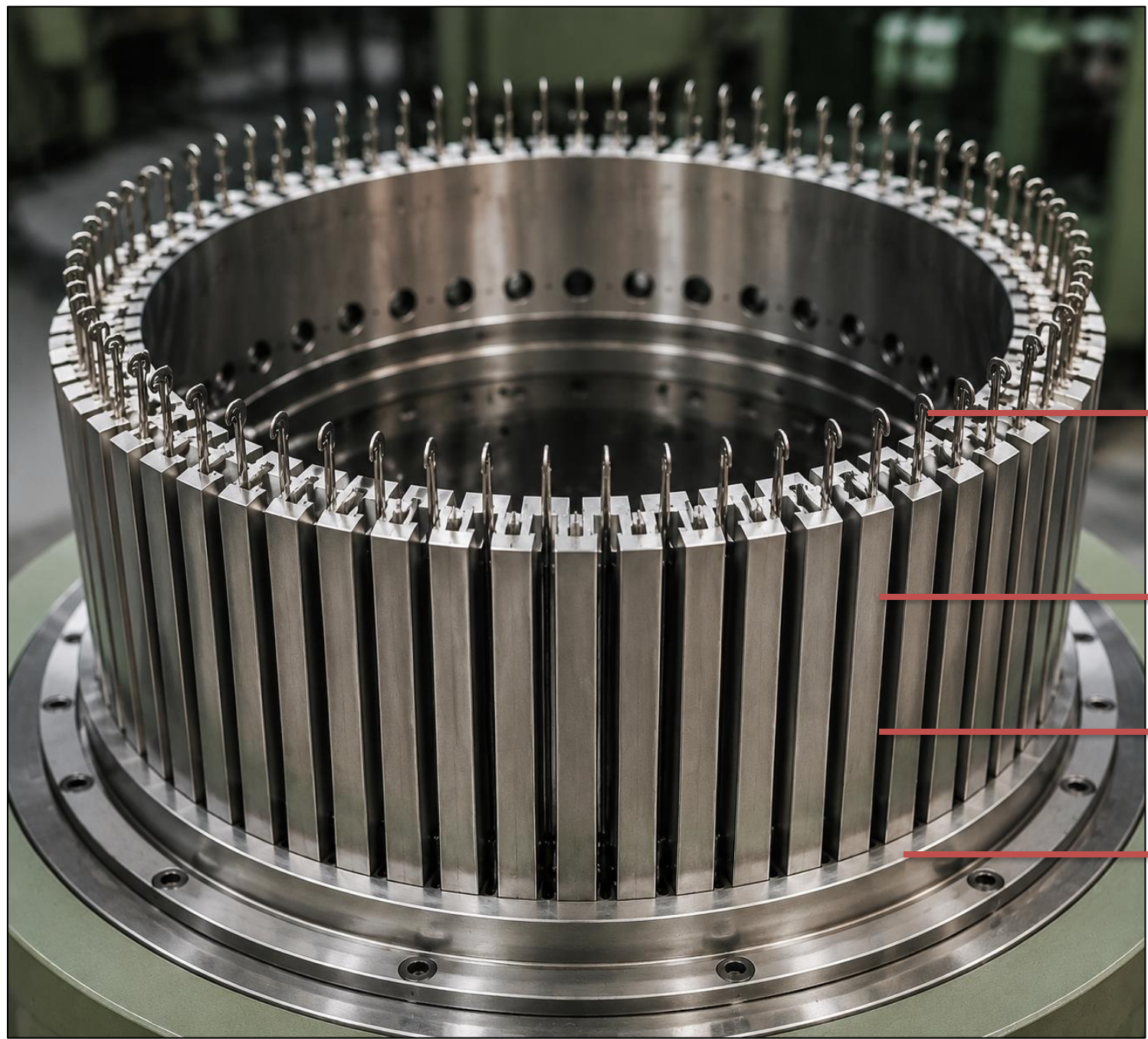
Independent needle



United needle



MACHINE PARTS



Needle

Trick Wall / Verge

Trick

Cylinder

BASIC TYPES OF KNITTED FABRIC CONSTRUCTION

RL Fabric (Right-Left Fabric)

- ❖ Also called *Single Face Fabric*.
- ❖ Produced using *one row of needles*.
- ❖ One side of the fabric shows right loops and the other side shows left loops.
- ❖ Example: Single jersey fabric.

RR Fabric (Right-Right Fabric)

- ❖ Also called *Double Face Fabric or Double Right Fabric*.
- ❖ Produced using *two rows of needles*.
- ❖ Both sides of the fabric show right loops.
- ❖ Example: Interlock fabric.

LL Fabric (Left-Left Fabric)

- ❖ Also called *Links-Links Fabric or Double Left Fabric*.
- ❖ Produced *using one or two rows of needles*.
- ❖ Both sides of the fabric show left loops.
- ❖ Example: Purl fabric.

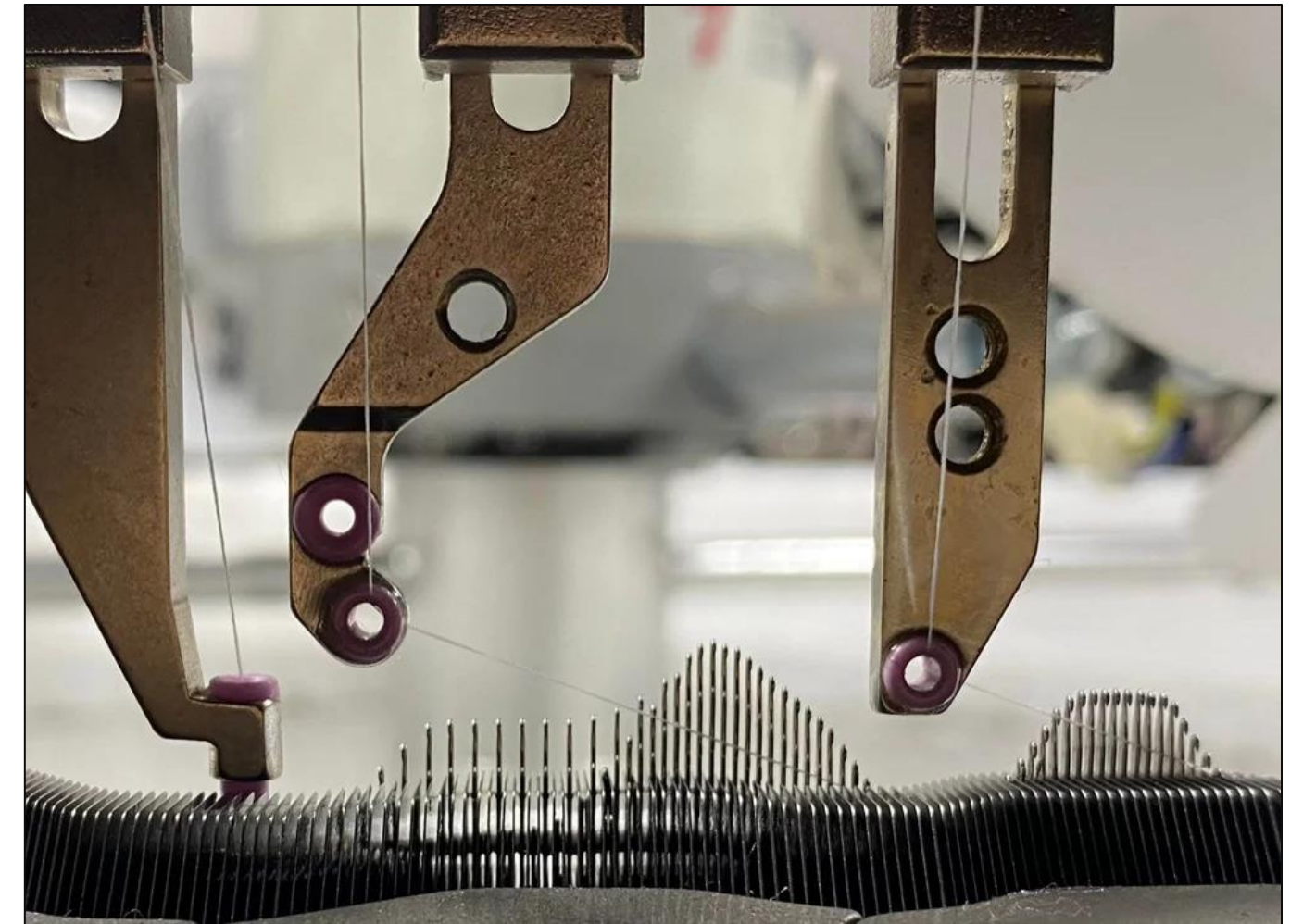
YARN FEEDER

Feeders are the yarn guides placed close to the needles to the full circumference of the knitting zone.

The feeders **feed the yarn into the needle hooks** and control the needle latches in their open position while the needle attain their clearing position.

On the circumference of a circular knitting machine up to 120 knitting cam sets can be mounted, each cam set fed with a separate yarn. This results in obtaining 120 knitted loop courses in one machine revolution.

NO OF FEEDER = NO OF CAM BOX = NO OF COURSE



GAUGE, PITCH

Needle Gauge

A measure expressing the *number of needle per unit of needle bed width (inch)*. It is denoted by G or E.

Gauge (E) = No of needles/One English Inch

Highest Gauge: 60. Lowest Gauge: 2 ~ 2.5

Pitch

Distance between one needle to neighbor needle. Pitch = $1/N$.

NO OF NEEDLES = NO OF WALES

SELVEDGE FABRIC

A selvedge fabric is one having a “self edge” to it and can only be produced on machines whose yarn reciprocates backwards and forwards across the needle bed so that a selvedge is formed as the yarn rises up to the next course at the edge of the fabric.

Selvedge fabric is primarily produced in flat-bed knitting machine. For knitted fabric, selvedge is parallel to the wales and welt is parallel to the course.



TUBULAR FABRIC

Produced directly as *a tube on circular knitting machines*.

In this fabric, Edges are not cut. Slitting mark may present but unlike to the cut edge fabric slitting is not carried out during knitting phase.

Rather it is done by slitting machine at a later phase.



CUT EDGE FABRIC

Cut edge fabric is usually produced by slitting open a tube fabric by an embedded blade in a circular machine.

A slit tube of fabric from a 30" diameter machine will have an open width of 94"(πD) at knitting and before relaxation.

Generally, needle is not placed in a particular track of a cylinder which acts a slitting mark. This kind of fabric is also called needle open fabric.



SLITTING MARK

Knit fabrics in circular knitting machine are produced in tubular form. For open width finishing line, this tube needs to be slitted open.

A slitting machine is used in between hydro-extractor and dryer. The machine finds out a line of absent needle in the fabric automatically and slits the fabric along that line.

Therefore, for open width finishing line, one or two needles are taken off intentionally from the knitting machine for generating slitting mark.



LAPS

Open Lap

Underlap is in the same direction as overlap or absent. It produces a lighter and more extensible structure.

Closed lap

Underlap returns in the opposite direction. It produces a heavier, denser, more opaque, and less extensible structure.

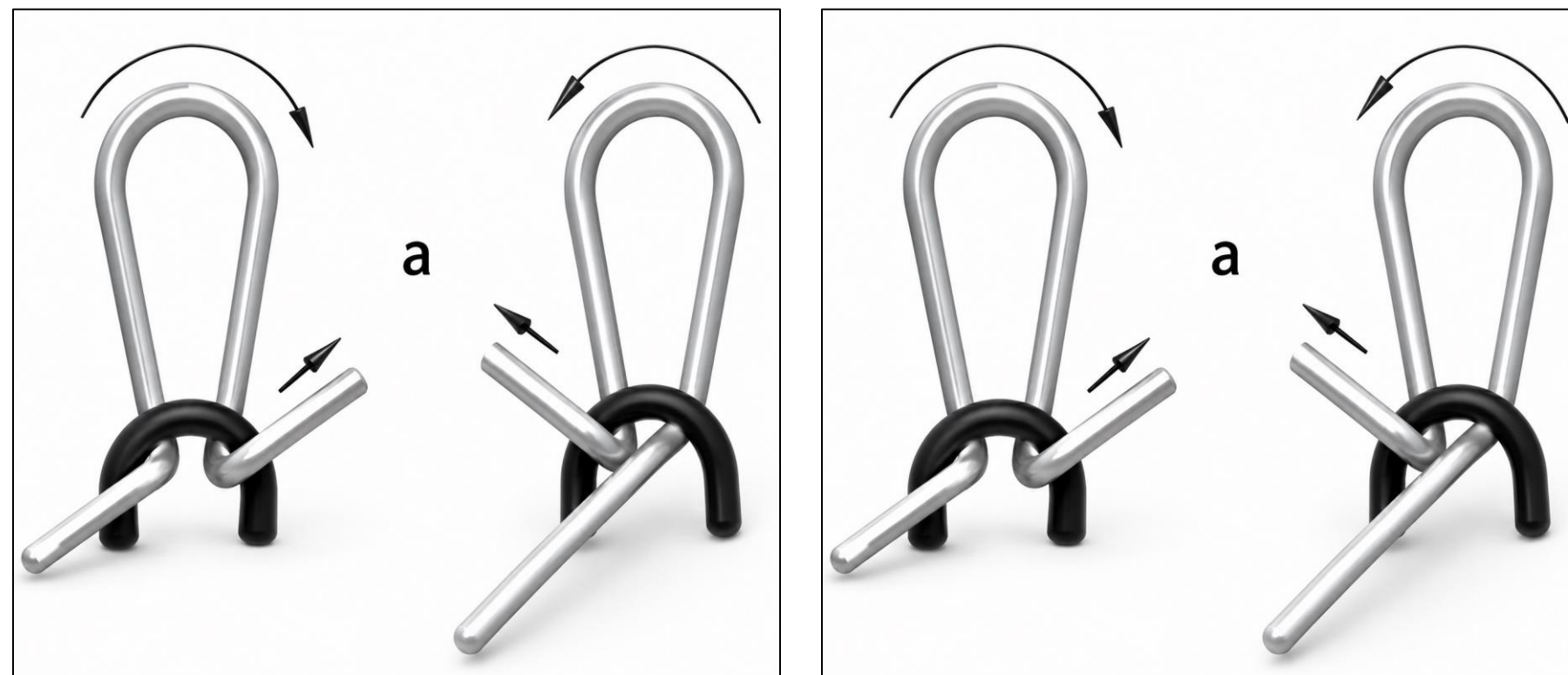
Overlap

In warp knitting, overlap and underlap are two important movements of the warp yarn. The overlap is the part of the yarn that moves across the needle hook and enters the needle to form a loop. The movement of the guide towards the hook and back creates the two sides of the loop. It looks similar to a needle loop in weft knitting. In simple words, overlap is the yarn placed over the needle during loop formation.

LAPS

Underlap

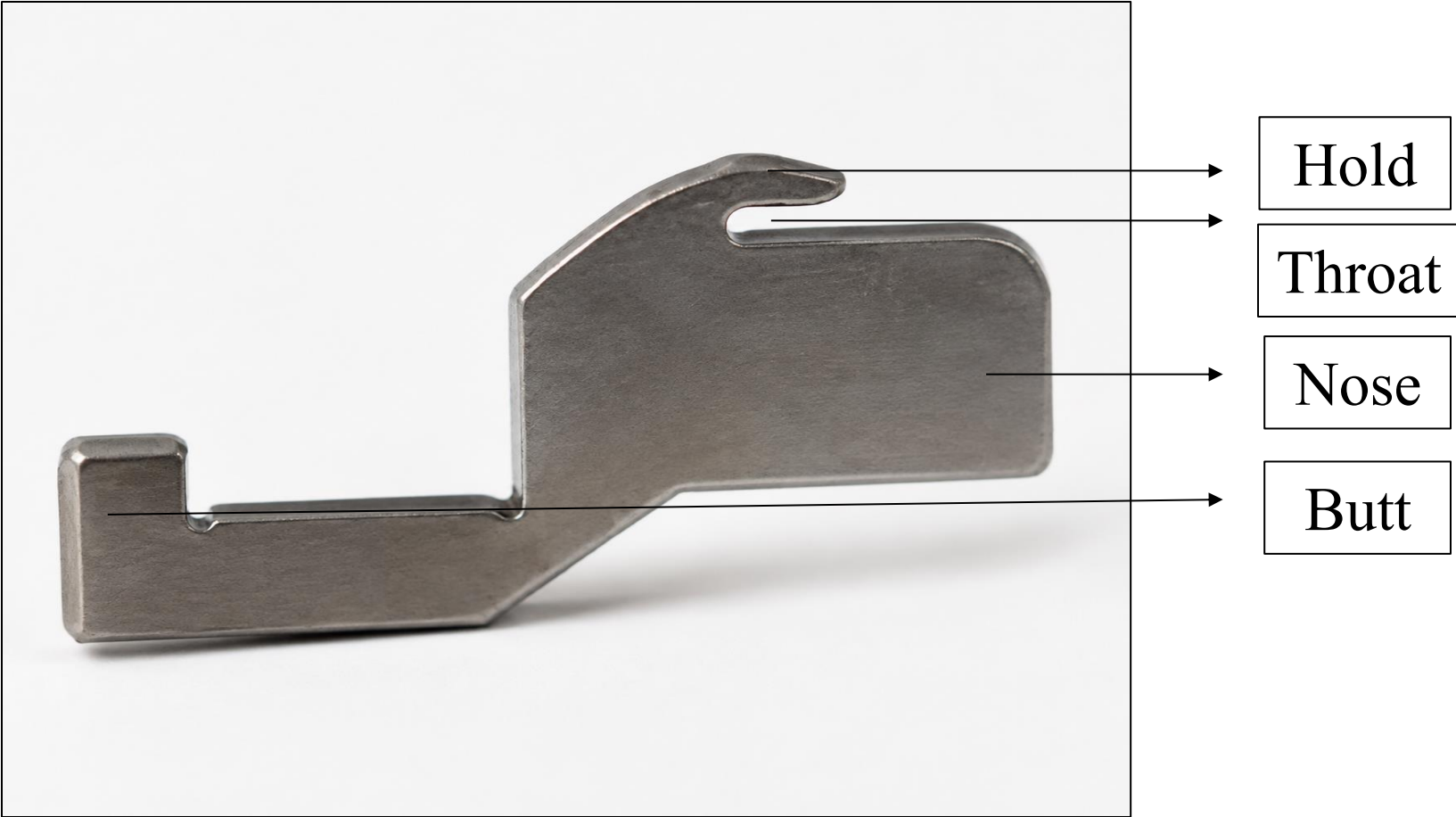
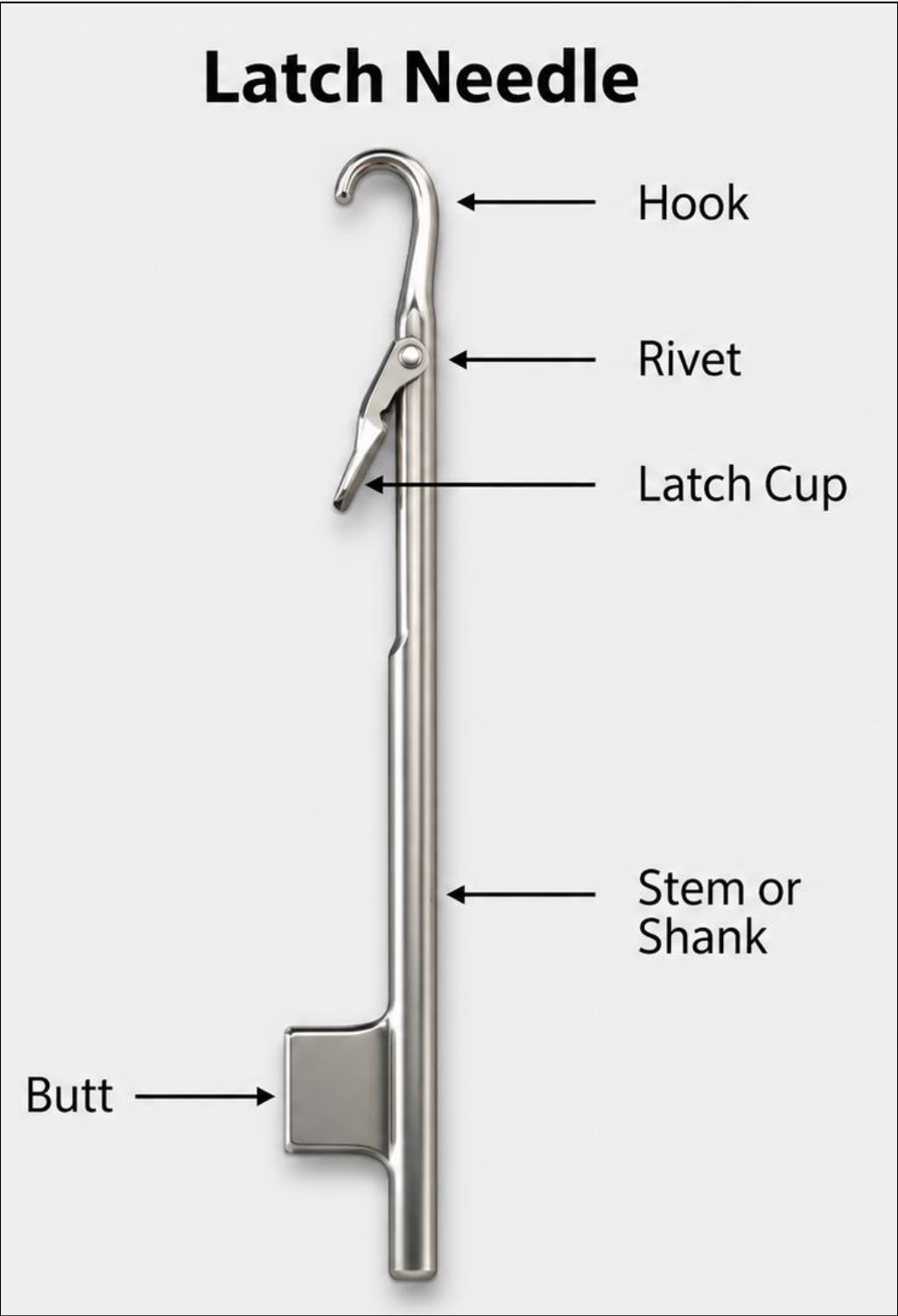
The underlap is the part of the yarn that moves on the opposite side of the needle hook. It connects one loop with the next loop in the following course. In simple words, underlap is the yarn that joins two consecutive overlap loops together. Both overlap and underlap are necessary in warp knitting because they connect the loops and help to form a stable fabric structure.



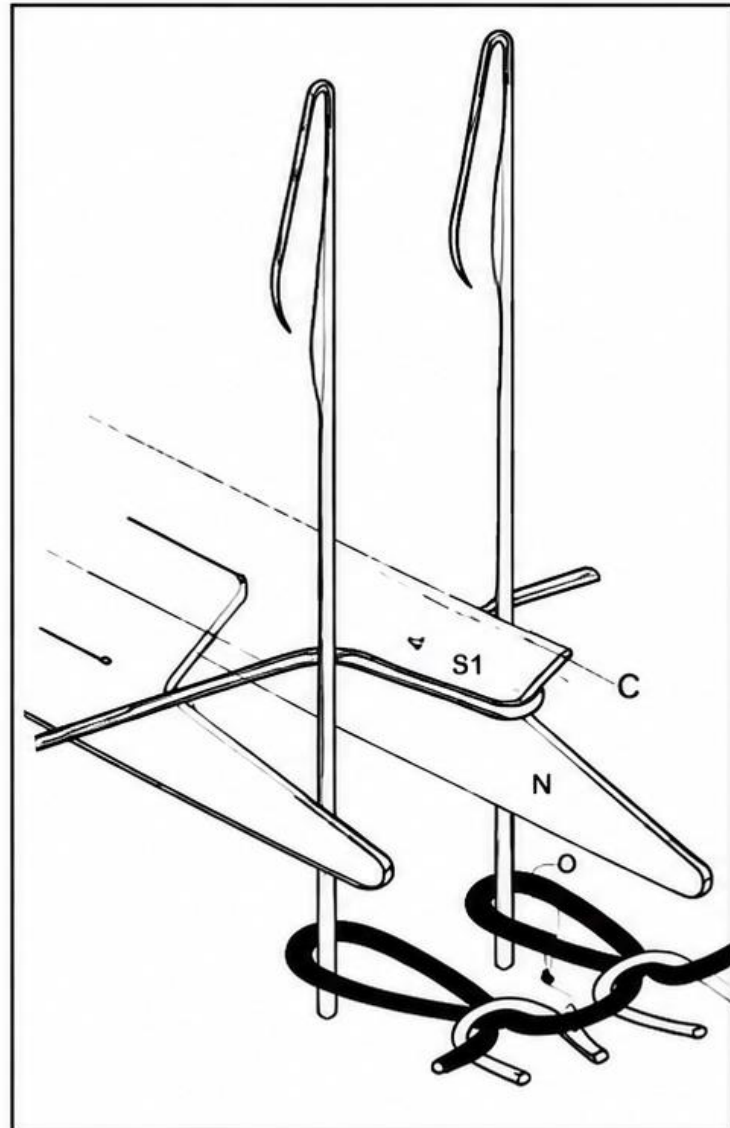
MATHEMATICAL FORMULA

1. No. of Needles = No. of Wales = $\pi \times \text{Machine Diameter (D)} \times \text{Machine Gauge (E)}$
2. No. Feeders = No. of cam-box = No. of Courses/rev.
3. Courses Per Day (CPD) = RPM (n) x Feeder (F) x Efficiency (η) x Time (60 x 24)
4. Fabric Length (m/Day) =
$$\frac{\text{Courses Per Day}}{\text{Courses Per Centimeter (CPC)} \times 100} = \frac{nF\eta t}{\text{CPC} \times 100}$$
5. Fabric Width (m) =
$$\frac{\text{No of Wales}}{\text{Wales Per Centimeter (WPC)} \times 100} = \frac{\pi DE}{\text{WPC} \times 100}$$
6. GSM =
$$\frac{\text{WPC} \times \text{CPC} \times 100 \times 100 \times \text{SL} \times \text{Tex}}{1000 \times 1000}$$
7. Production (Kg/Day) =
$$\frac{\text{CPD} \times \text{CL} \times \text{Tex}}{1000 \times 1000 \times 1000}$$
8. Course Length (CL) = Stitch Length (SL) x π DE

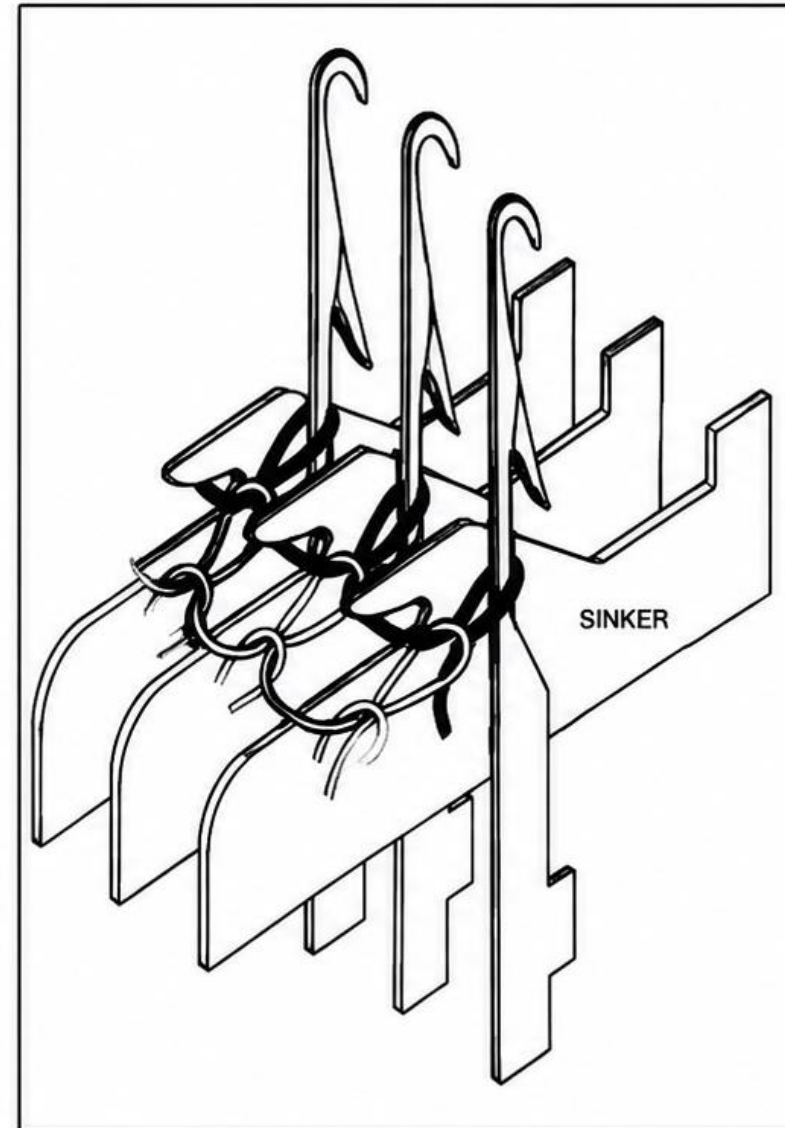
KNITTING ELEMENTS



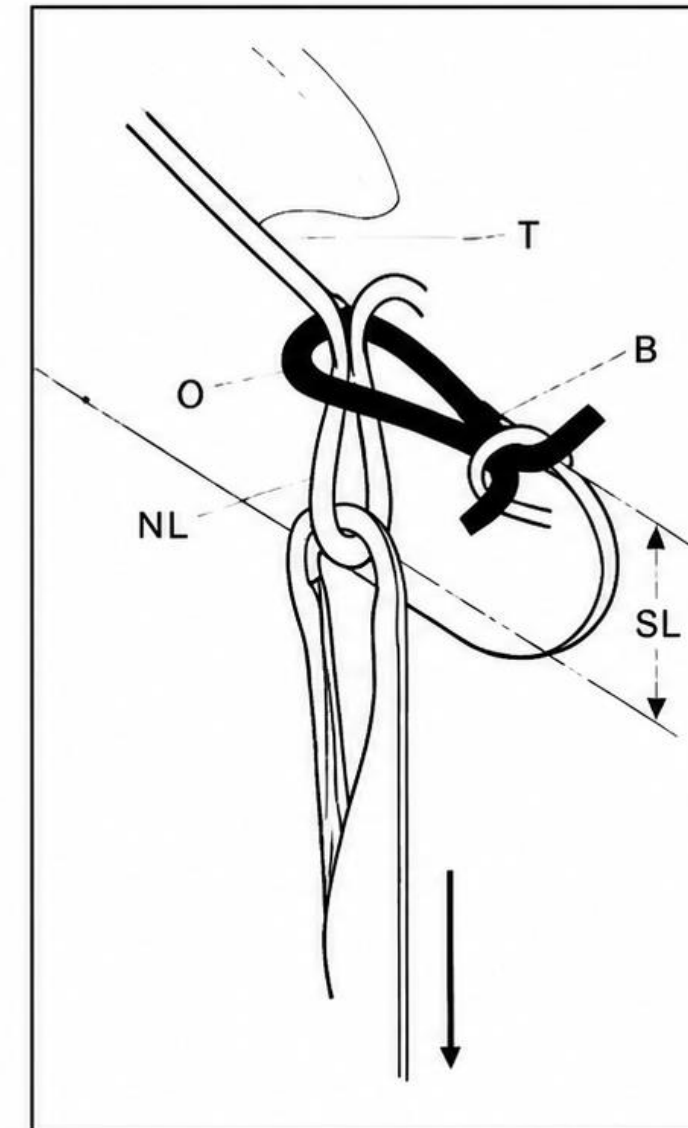
KNITTING ELEMENTS



Loop forming Sinker



Holding – down Sinker



Knocking – over Sinker

KNITTING ELEMENTS



Knit Cam

Tuck Cam

Miss Cam

THANK
YOU !

